



Flexibility costs of debt: Danish exporters during the cartoon crisis

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ARTICLE INFO

Article history:

Received 7 July 2021

Revised 3 March 2023

Accepted 4 March 2023

Available online 20 March 2023

JEL classification:

F14

G31

G32

Keywords:

Flexibility

Capital structure

Product markets

Consumer boycott

ABSTRACT

We study how firms respond to an unexpected demand shock, exploiting the 2006 boycott of Danish products after publication of Muhammad caricatures. On average, affected firms lose the majority of their exports to Muslim countries and experience a significant decrease in total sales. However, firms with low financial leverage redirect sales to new and existing product-destination markets in non-Muslim countries, which allows them to fully offset their losses. In contrast, high-leverage firms do not enter new markets and instead actively downsize. Our results highlight the importance of financial flexibility in times of crisis, consistent with declarations of practitioners.

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1. Introduction

Resilience to shocks and the ability to adapt to a new environment are key characteristics of successful firms. The importance of adaptation is evident in the world of increasing business uncertainty,¹ and is confirmed by declarations of practitioners, for whom flexibility tops the list of factors that drive their financial decisions.² Among various types of shocks that affect firms, changes in their product

market environment are a particularly common and important source of uncertainty, and hence it is crucial to understand how firms can successfully adapt to them.

When a negative product market shock lowers demand for a firm's products, the firm may try to recoup lost revenues by adjusting prices or by changing what and where they are selling. While several existing studies have explored price adjustments (Chevalier, 1995; Zingales, 1998; Gilchrist et al., 2017; Driscoll, 2018; Ge, 2022), evidence on changes in the set of destinations that the firm serves or the set of products that it sells in these locations is scarce. These market adjustments are rarely observed in the data, in particular in conjunction with a shock that triggers these adjustments and allows for their credible analysis.

In this paper, we analyze an unexpected and idiosyncratic demand shock, studying how firms respond to it by entering new markets and how they complement these choices with other adjustments. We take advantage of the 2006 Cartoon Crisis, a widespread boycott of Danish

The code for this article can be found at <https://doi.org/10.17632/w9mg2vmz3z.1>

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¹ Baker et al. (2016) provide systematic evidence on the recent increase in economic uncertainty.

² Among 392 CFOs surveyed by Graham and Harvey (2001), 59.4% say that financial flexibility is important in affecting how they choose the appropriate amount of debt for their firms. This is the highest value among all 14 factors listed, including factors such as taxes or bankruptcy costs.

products in Muslim countries after a Danish newspaper published caricatures of prophet Muhammad, and utilize uniquely rich administrative data on firms' balance sheets and product-level exports. Our main findings show that firms can neutralize the boycott-induced drop in sales by taking advantage of sales opportunities elsewhere, provided they have enough financial slack beforehand. Low-leverage firms enter new markets and adjust their product portfolio, whereas high-leverage firms show no signs of expansion and instead respond by downsizing and cutting labor costs. Our analysis demonstrates the link between firms' financial positions and their ability to adjust to shocks, in particular via new market entry, which is new to the literature. As such, our results provide rare direct evidence on the nature of the flexibility costs of debt often reported by practitioners (Graham and Harvey, 2001).

We start by describing the course of the boycott. In September 2005, a Danish newspaper published caricatures of prophet Muhammad, which is considered blasphemous by Muslims. The publication led to a widespread boycott of Danish products in Muslim countries, which affected Danish exporters for at least one year. Exposed firms lost the majority of their exports to Muslim markets, which for an average firm should translate to ~6% decline in total sales. Yet, our analysis reveals that sales of affected firms only drop by 3% on average. This muted response suggests that firms increase their sales elsewhere, which is confirmed by a ~16% increase in exports to non-Muslim markets. At the same time, we document a reduction in costs, as evidenced by a small, yet significant decline in total wages.

Coexistence of these various ways of adjustment motivates us to analyze the heterogeneity in the response to the boycott across several dimensions. Perhaps surprisingly, we find no significant heterogeneity in adjustment by firm size or manager compensation, and only limited differences between firms with different breadth of pre-boycott product portfolio.³ The most striking finding, however, is related to heterogeneity by firms' financial leverage, defined as the share of total liabilities in total assets. Low-leverage firms successfully recoup the entire lost demand, while firms with high leverage do not significantly expand, lose over 6% of total sales, and reduce employment.

What explains the significant heterogeneity by firms' capital structure? We document that after the boycott low-leverage firms enter significantly more new markets, which we define as product-country pairs in which the firm had no sales before. These new markets represent both entry into new non-Muslim countries and the expansion of the set of exported products sold in previously-served countries. The market entry response is muted for high-leverage firms, and the total number of markets they serve decreases.

What are common characteristics of new destination countries that low-leverage firms enter? Our analysis reveals that market entry is most pronounced in countries

where firms' own export growth in the past was relatively low, but where other Danish firms have been widely present and were growing quickly. Hence, firms seem to identify markets with untapped potential for their own products, which at the same time have already proved to be welcoming to other Danish firms. The new destinations also tend to be richer and linguistically more similar to Denmark than a typical country, although they are spread around the whole world. High-leverage firms also expand, but only into few selected types of markets that appear to be most lucrative and cheapest to enter.

Market entry also takes the form of expanding a firm's product portfolio in countries that the firm has already served. The most natural form of this response is product redirection: we document that firms enter non-Muslim markets with the same products that they just stopped selling to Muslim markets. This response is larger for low-leverage firms, but high-leverage firms also display a smaller, but positive and significant effect along that margin. Selling the same product elsewhere is likely the easiest and cheapest response, but oftentimes firms must adjust their production to better serve new customers. Indeed, we demonstrate that low-leverage firms expand their product portfolio in a wide set of product groups, both those with and without a strong presence in the past. In sum, the number of export products significantly increases for low-leverage firms. The response of high-leverage firms, however, is muted, and their number of exported products remains unchanged.

Analyzing a subset of firms for which we also observe product-level domestic sales, we can demonstrate that a smaller part of the market entry response is driven by new additions to the firm's product portfolio. This adjustment, which again is only significant for low-leverage firms, happens quickly, suggesting that new products are typically closely related to products sold before the boycott, which are redesigned, repackaged, or adjusted to better serve the new markets. Market entry and product portfolio adjustment are generally larger for fast-moving industries and product groups with higher recent entry rates, consistent with these settings offering more room for product adjustments and suggesting that some changes may reflect acceleration of planned updates in the product portfolio in response to the boycott.

Our findings highlight the importance of flexibility costs of debt, even if the firm is not close to bankruptcy. The finding that only firms with higher level of financial slack are able to adjust to changing conditions by expanding into new markets is consistent with the mechanism of financial constraints: an adverse economic shock combined with high level of debt limits firms' ability to finance costly and risky expansion and increases their emphasis on short-term profits (Chevalier and Scharfstein, 1996; Stein, 2003). At the same time, it is at odds with other theories of how debt affects firms' adaptation.⁴ Consistent with the impor-

³ Firms with a larger product portfolio are more likely to enter new markets after the shock, consistent with basic economic intuition, but these new destinations account for a small share of their business and cannot mitigate the sales decline.

⁴ Debt may increase the urgency of adaptation and disincentivize taking a wait-and-see approach, especially for a shock with an uncertain duration: high-debt firms are at higher risk of bankruptcy if they do nothing, and hence may have stronger incentives to actively respond to the revenue shortfall (Jensen, 1986). At the same time, pre-commitment to debt

tance of financing the expenditures on fixed costs of market entry, marketing, and adjustment of production capacity, we find a significant increase in SG&A expenses, investment, and debt volume among low-leverage firms, but no change for firms with high leverage. Exploring the behavior of labor costs, we document that high-leverage firms see a decline in employment and shift towards more flexible labor contracts, i.e., increase the use of outsourcing. These patterns again highlight the importance of flexibility, as they suggest that high-leverage firms might try to compensate for their lack of financial flexibility by increasing operational flexibility. These results also indicate that both low- and high-leverage firms actively respond to the boycott but they do so along different margins, consistent with differential importance of financial constraints for various types of adjustment.

A natural concern about our interpretation of the results is that firms with different levels of leverage may behave differently not because of financial constraints, but because of other omitted variables that are correlated with leverage. While our empirical design with firm-fixed effects accounts for time-invariant differences across firms, some firm characteristics may matter in adjusting to a crisis. For example, low-leverage firms may have managers with higher skills, or a larger product portfolio that is better suited for responding to the boycott via expansion. Alternatively, high-leverage firms might have expanded more before the boycott, and hence have less opportunities to expand now, or might have faced lower volatility of pre-boycott sales in general, and hence are less prepared for adaptation. We use two strategies to address this concern. First, we explore a loan-level dataset that, in the spirit of Almeida et al. (2011), allows us to identify firms with a high share of long-term debt that happens to mature in the year after the boycott – plausibly exogenous variation in the need to raise financing, and thus in financing constraints. When we compare these debt-at-maturity firms to other firms hit by the boycott, we see the same patterns as in the response of high-leverage firms. While firms with a high share of debt at maturity lose the same share of exports to Muslim markets, they do not enter new non-Muslim markets and the total number of the markets they serve decreases, which is also accompanied by a decline in employment. Our second strategy controls directly for potential confounding factors and their role in adjusting to the boycott. Using our granular data on managers' pay and education, past sales and exports, factors of production, product portfolio composition, and numerous other sources of heterogeneity, we find no evidence that our results are driven by any of these alternative explanations.

Several existing papers in the corporate finance and product market literature analyze the link between firms' financial decisions and pricing (Chevalier, 1995; Dasgupta and Titman, 1998; Busse, 2002; Khanna and Tice, 2005; Gilchrist et al., 2017; Driscoll, 2018; Ge, 2022), or total output or sales (Phillips, 1995; Fresard, 2010). We extend this evidence by analyzing new market entry and product

variety, on top of exploring other related decisions, such as pricing, employment, investment, and financing decisions. The results reveal a meaningful difference between the impact of debt on various margins of adjustments: while highly indebted firms may be more likely to lower prices,⁵ we show that these firms are less likely to expand via new market entry. These two mechanisms of pricing and product-market portfolio adjustments imply opposite effects of debt on competition,⁶ but they can be reconciled by noting that price cutting does not require additional investment and may provide a short-term boost in revenues without short-term costs, while the opposite is likely true for new market entry. Part of the market entry response that we observe is due to an expansion of firms' product portfolios. While several existing studies explore the link between finance and innovative outcomes of the R&D process (see Kerr and Nanda, 2015 for a review of the literature focused on patents, and Krieger et al. (2022) and Granja and Moreira (2022) for other measures), our results highlight the importance of smaller and faster adjustments in the product portfolio that allow firms to enter new markets and adapt their product offerings to local demand.

The context of our analysis connects us also to the literature on corporate finance and international trade (see Foley and Manova, 2015 for a review). Access to financial capital has been shown to be important for exporting activities (Amiti and Weinstein, 2011; Manova, 2013). Paravisini et al. (2015) find that bank credit shocks in particular affect the intensive margin of exports within product-destination. At the same time, the ability to enter new markets has been shown to be an important determinant of adjustment to a local shock (Almunia et al., 2021). Our analysis shows that capital structure affects the flexibility to enter new markets with new products, in response to an adverse demand shock in a healthy and stable banking environment. The nature of our shock also relates us to a small literature studying the impact of political tensions on bilateral business relations and firm value. For example, Heilmann (2016) documents the effectiveness of several international trade boycotts (the US-French conflict over Iraq war 2003, the Danish Cartoon Crisis 2005, and Sino-Japanese tensions 2012) in reducing aggregate bilateral trade flows between affected countries.⁷ At the firm level, Fisman et al. (2014) show that recent Sino-Japanese tensions reduced firm value for firms with high exposure to the affected market. We take one step further and analyze how firms adjust to boycott-induced losses across product-market decisions and input choices,

⁵ Zingales (1998); Busse (2002) and Driscoll (2018) document that high-leverage firms are more likely to reduce prices, but Chevalier (1995) finds the opposite. In our setting, we do not find significant differences in pricing response of low- and high-leverage firms, but the insignificant patterns we document are consistent with high-leverage firms having lower willingness to reduce prices.

⁶ For various perspectives on the effect of debt on competition, see e.g., Brander and Lewis (1986); Maksimovic (1988); Kovenock and Phillips (1997); Dasgupta and Titman (1998); Khanna and Tice (2000); Campello (2003); Hadlock and Sonti (2012).

⁷ See also Michaels and Zhi (2010); Friedrich (2022), as well as Davis and Meunier (2011) and Tanaka et al. (2019) for more evidence on each of these trade episodes.

payments of a fixed size may make it optimal to respond by stabilizing revenues rather than cutting costs. We provide additional discussion of theoretical mechanisms in Section 6.1.

and how these adjustments depend on firm capital structure.

Our findings about the link between capital structure and firm's ability to adapt to a shock extend the evidence from several studies that analyze flexibility in pricing (D'Acunto et al., 2018), production technology (Reinartz and Schmid, 2016), employment contracts (Simintzi et al., 2014; Serfling, 2016; Kuzmina, 2022), and the general link between financial and operational flexibility.⁸ While these papers study the equilibrium choice of capital structure and its relation to various aspects of flexibility, we present direct evidence on how the capital structure decisions made before an exogenous shock influence the adaptation to that shock. Our approach is similar to that taken by Giroud and Mueller (2017), who show that high leverage firms suffered larger employment losses during the great recession. We find an employment response that is consistent with their results, but our focus is on market entry response and our setting features a sudden and highly unexpected shock that had a significant impact on the demand of a selected group of firms, but did not influence the health of the financial sector or labor market conditions.⁹ More broadly, we contribute to a larger literature on the impact of capital structure and financial constraints on real outcomes.¹⁰ Our setting allows the analysis of multiple outcomes in one dataset. The analysis of our main outcome – new market entry – is novel to the literature, while other margins have received significant attention earlier. Yet, our advantage is being able to analyze them in one dataset, compare their relative importance, and understand how different margins of adjustment complement or substitute each other.¹¹

⁸ Various ways of relating operational and financial flexibility have been proposed in the literature (Novy-Marx, 2011; Gu et al., 2018; Chen et al., 2019). Our results on outsourcing confirm a seminal theory of how operational and financial flexibility are linked – the trade-off hypothesis (Van Horne, 1977) – in a quasi-experimental setting.

⁹ Our shock is also a useful laboratory for studying the effects of demand shocks in general, which the literature often analyzes as a part of broader events (e.g., currency depreciations, Desai et al. (2008)), leading to difficulties in separating the effects of the demand shock from other factors.

¹⁰ Most existing studies analyze financial shocks, e.g., shocks to credit supply (Khawaja and Mian, 2008; Chava and Purnanandam, 2011; Nanda and Nicholas, 2014), or cash windfalls (Lamont, 1997). Particularly related is recent paper by Beaumont and Lenoir (2019), which studies how a liquidity shock impacts a firm's ability to expand their customer base. We take a different perspective, analyzing a real shock that tightens financial constraints, and study how financial choices made by the firm before the shock affect the adaptation. This approach is similar to Ofek (1993); Giroud and Mueller (2017), and Kini et al. (2017), and enriches findings from studies of financial shocks by reflecting changes in credit demand, not supply (Amiti and Weinstein, 2018).

¹¹ E.g., Chodorow-Reich (2014); Falato and Liang (2016); Baghai et al. (2021); Caggese et al. (2019) study various aspects of personnel decisions, Chava and Roberts (2008) study investment. We extend this evidence and show that labor hoarding among low leverage firms may be facilitated by better ability to sustain revenues, and that the employment decline for high leverage firms may be accompanied by an increase in outsourcing. Moon and Phillips (2021) and Kanatas and Qi (2016) argue that firms that outsource have lower financial leverage to protect relationship-specific investments. Our paper suggests an alternative mechanism that works in the opposite direction: firms with high leverage choose to outsource to increase operational flexibility.

2. Cartoon crisis

Background. Denmark's largest newspaper, *Jyllands-Posten*, published 12 cartoons of the prophet Muhammad on September 30, 2005. According to the newspaper's accompanying article, the cartoons were a statement in favor of freedom of expression, in response to the self-censorship of Danish artists regarding illustrations in a recently published book about the life of Muhammad.

The cartoons first led to public protests among Danish Muslims that received no formal response from *Jyllands-Posten* or the Danish government. As a consequence, a group of Danish Muslims contacted ambassadors of several Muslim countries to Denmark for help in disseminating information about the cartoons in the Muslim world. The group was successful in placing the cartoons on the agenda of the Organization of Islamic Countries conference in Mecca in December 2005. This event set in motion widespread media coverage and political debate in Muslim countries. By the end of January 2006, Saudi Arabia and Kuwait were the first countries to declare an official boycott of all Danish products. These announcements were followed by more violent protests at Danish embassies in Syria, Iran, Pakistan, and other countries. At the same time, the boycott quickly spread to Muslim countries around the world. For a detailed time line of events, see Jensen (2008).

To illustrate the actual timing and duration of the boycott, the left panel of Fig. 1 compares total monthly exports from Denmark and from the rest of the EU to countries with at least 50% Muslim population.¹² The horizontal axis defines time (in months) relative to September 2005 when the cartoons were first published. As the timeline illustrates, the boycott started with a delay of several months because the dissemination in the Muslim world took a considerable amount of time. The figure shows that while exports from other European countries continued to grow over this period at their previous rate of around 10%, Danish exports experienced a sudden drop and remained at 25% below their previous level for more than one year. Danish exports largely recovered in 2007 at the aggregate level after the conclusion of the boycott.

The aggregate time series suggests a temporary shock with a full recovery by mid-2007 compared to the export volume from other EU countries. But this aggregate time series hides important heterogeneity in the persistence of the shock across Danish exporters. The right panel of Fig. 1 shows exports to Muslim countries for firms that shipped products to these markets in each year between

¹² Consistent with the findings by Michaels and Zhi (2010), a deterioration in attitudes towards Danish products led to a substantial reduction in Danish exports even to Muslim countries that did not declare an official boycott. Muslim population shares follow a report by the Pew Research Center (2009) based on national census data from the years 2000–2006 and the World Religion Database using Muslim population estimates for the year 2005. The countries are United Arab Emirates, Afghanistan, Albania, Algeria, Azerbaijan, Bangladesh, Burkina Faso, Bahrain, Brunei, Djibouti, Egypt, Western Sahara, Gambia, Guinea, Indonesia, Iraq, Iran, Jordan, Kyrgyzstan, Comoros, Kuwait, Kazakhstan, Lebanon, Libya, Morocco, Mali, Mauritania, Maldives, Malaysia, Niger, Nigeria, Oman, Pakistan, Palestine, Qatar, Saudi Arabia, Sudan, Sierra Leone, Senegal, Somalia, Syria, Chad, Tajikistan, Tunisia, Turkey, Uzbekistan, Yemen, and Mayotte.

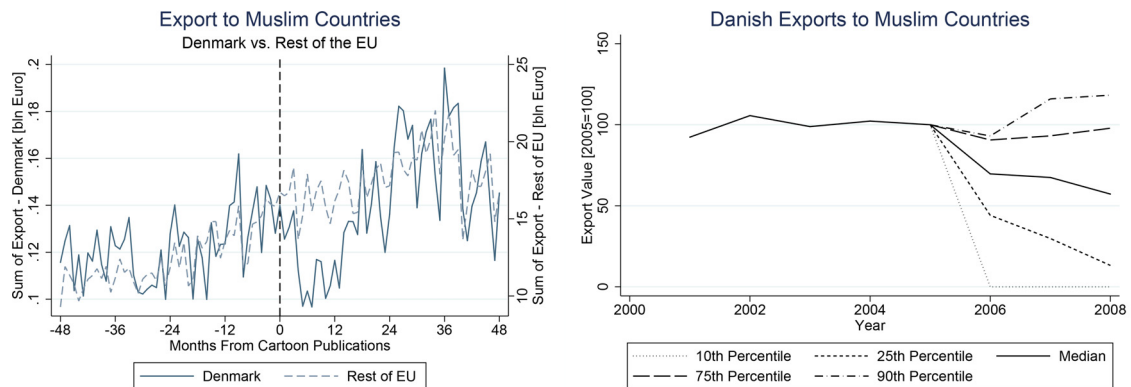


Fig. 1. The evolution of exports to Muslim countries. The left panel shows total monthly exports from Denmark and from the rest of the European Union to countries with at least 50% Muslim population. The X axis defines time (in months) relative to September 2005, the time of cartoons publication. The data comes from Eurostat. The right panel shows firm-level evolution of exports to Muslim countries over time for Danish firms with positive exports to Muslim countries in each year over 2002–2005. The value of firms' exports in 2005 is normalized to 100 and the percentiles are calculated based on the distribution of exports in each respective year. These data are provided by Statistics Denmark.

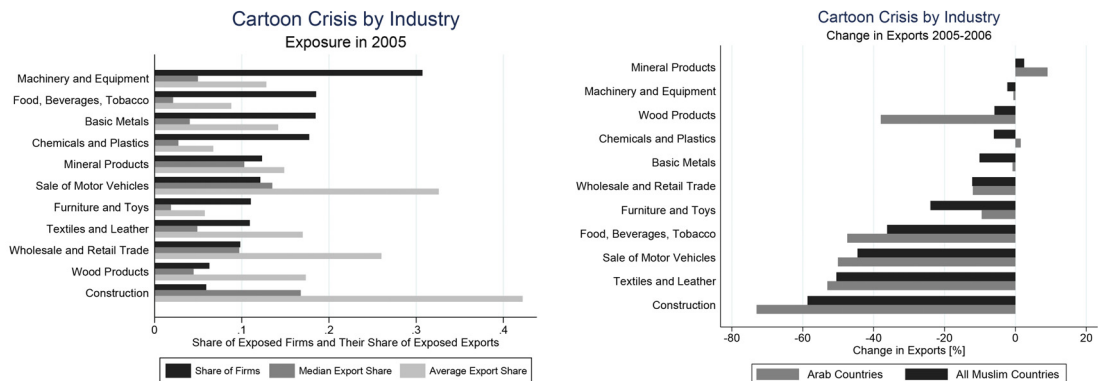


Fig. 2. Boycott across Industries. The left panel shows the extensive and intensive margin of exposure: share of firms who were exposed to Muslim countries by industry (where exposure is defined as having at least 0.5% of total exports sold to Muslim countries) as well as mean and median share of exports to Muslim countries among exposed firms by industry. The right panel shows the change in export by industry for all Muslim countries and for Arab countries.

2002–2005. Since exports are naturally volatile and in each year some previous exporters lose their presence in foreign markets, we choose a stable sample to provide a more conservative visualization of the boycott effect. Yet, even among firms with such long-standing export relationships to affected markets, the effect of the boycott is dramatic. At least 10 percent of these firms persistently lost their entire presence in Muslim countries, and effects for the median firm show a substantial export decline which these firms were unable to recover in subsequent years. In contrast, the aggregate recovery in Fig. 1 is driven by high-growth firms and by new entrants into these markets. In addition, it is important to remember that firms faced substantial uncertainty ex-ante about the duration and severity of the boycott, and their expectations might have been different than the actual realization of the events.

The left panel of Fig. 2 illustrates extensive and intensive margins of Danish firms' initial exposure. Exposed

firms constitute a substantial fraction of exporters in a large set of different industries, ranging from consumer products such as textiles, food, and furniture, to heavy machinery and equipment. Among exposed firms, there was considerable variation in the importance of Muslim destinations not only across sectors but also across firms within industries. The difference between average and median shares by industry indicates a small share of firms with high exposure in each industry. Consistent with that, Appendix Figure A.1 shows the density of firm-level exposure. While most firms have relatively small exposure, some are focused almost exclusively on Muslim markets. The right panel shows the decrease in exports by industry to all Muslim countries and to Arab countries separately. Construction and consumer products experienced the largest drop in exports. Even though the boycott was most salient in the Arab countries, observed export changes are similar as those for Muslim countries as a whole.

Discussion. How could firms adjust to this sudden and unexpected boycott in some destination countries? First, the boycott is a negative shock to firm revenues, but at the same time may change the valuation of other potential projects faced by the firm. Serving other non-Muslim markets possibly had negative net present value (NPV) before the boycott, but because of sunk costs, existing inventory, and idle capacity, it may have positive NPV after the boycott. Yet, redirecting sales may be costly, because of fixed costs of foreign market entry (e.g., to establish business contacts, register their sales, see [Eaton et al., 2011](#)) and because even selling additional products in pre-established destination countries requires expenditures on product repackaging, product modifications, and increased sales efforts.

Second, revenues may also be recovered by changing prices, even if the firm was pricing optimally before the boycott. For example, with higher short-term pressure to sustain revenues, the firm may decide to increase prices if the short-term elasticity of demand is below 1, sacrificing long-term market share in favor of higher revenues today. On the other hand, if the elasticity of demand is above 1, firms may decide to lower prices to generate more revenues. Pre-boycott, this higher level of revenues might have been suboptimal from a profit perspective, but post-boycott, with several cost categories being sunk, a low-price strategy may be more appropriate.

Finally, firms may respond by cutting costs. For example, firms may stop ordering raw materials needed for goods that used to be sold in Muslim countries but will not be produced anymore. For other cost categories, however, a cost-cutting strategy may provide savings in the short-run but may generate even higher costs in the long run. For example, if a firm fires highly skilled and specialized workers, this approach may reduce current labor costs, but may lead to even higher recruitment and training costs in the future, when the need for these workers arises again. This possibility is particularly relevant if the firm expects the boycott to be short-lived. The optimality and ability to choose between these strategies may depend on firm characteristics.

The level of the expansion costs and ability to pay them may vary across firms, for example because of differences in their product portfolio or existing distribution network. Firms may differ in the share of products marketed specifically to Muslim markets, in expertise needed to expand, or in their ability to obtain capital needed for financing the expansion, even if it has positive net present value. Similarly, a firm's financial condition may affect its employment strategy because sustaining significant labor costs during a period of low revenues may be infeasible for firms who do not have a large financial buffer and cannot easily borrow against future earnings. Overall, firms' choices are informed by their capabilities and constraints, and will be reflected in changes in market entry and product portfolio adjustment on the one hand, and sales and investment expenditures, as well as the level of various cost categories, on the other hand. Examination of changes in these outcomes is the goal of our empirical analysis.

3. Data and methodology

3.1. Data

We combine several administrative data sources for the universe of firms and workers in Denmark, including firm-level trade data, financial statements of firms, and employer–employee matched data. In addition, for a sample of firms, we are able to add detailed information on product-level total sales, as well as outsourcing and input expenditures. The baseline period of our analysis is 2001–2006. We provide selected results in an extended sample ending in 2010, but refrain from using the longer time frame in our main analysis for several reasons: changes in the product classification in 2007 that may impede our ability to credibly analyze the product-market response; the onset of the financial crisis in 2008; and the duration of the boycott being limited to roughly one year.

The first data source is the Danish Foreign Trade Statistics Register (UHDI), which provides annual firm-level trade value by product-destination pairs at the CN-8-digit product level. Importantly, any trade flows with countries outside the EU are precisely measured by the customs authority (Extrastat). For trade with EU member states, Danish firms only have to declare exports above a threshold of approximately \$250,000 per year (Intrastat). Thus, we do not include firms selling small quantities only to destinations within the EU in the sample of exporters. In addition, below-threshold exports to European Union countries will not be reflected in our total exports figures, but they will be reflected in total sales figures. In other words, small exports to EU countries are recorded as domestic sales.

Second, we add financial statements of firms from the Accounting Register (FIRE) and additional information on founding date, sales, employment, industry, and firm exit from the Danish Business Register (FIRM). The accounting data provides balance sheets and profit and loss statements with information about short-term and long-term liabilities, assets, investment, and input costs. A smaller subset of manufacturing firms also provides more detailed responses on purchases of goods and services, with a specific section of the survey listing expenditures on outsourcing across different tasks, such as transportation, accounting, consulting, catering, IT, and marketing. Moreover, a similar dataset collects information on sales by product CN-8-digit code for all manufacturing firms with at least 10 employees.¹³ These data will be valuable to identify adjustments of the product portfolio among these firms.

Third, we utilize loan-level data on secured and unsecured loans (excluding mortgages) issued to Danish firms by Danish banks. We observe the amount of the loan, the interest rate, and the moment when the loan was repaid, but we do not observe the original maturity at origination. Therefore, we identify long-term debt by dropping loans that only appear in the data for one year. We use this data to calculate the share of long-term debt that is maturing

¹³ [Smeets and Warzynski \(2013\)](#) provide more details about this dataset and use it to measure productivity of multi-product firms.

Table 1

Selected summary statistics.

We report the sample median sales; employment (number of full-time equivalent workers); investment (capital expenditures); sales, general & administrative expenditures; and leverage (total liabilities divided by total assets) and sample means for export shares; number of export destinations; export products; and total new products. The number of observations is for the full dataset. We note that the number of new products is only available for a subsample of manufacturing firms. Monetary values are in USD.

	Exposed	Control	Total
Firms	1687	13,213	14,900
Observations	7515	53,861	61,376
Sales	4.4 M	1.6 M	1.8 M
Employment	21.4	8.0	8.8
Investment	0.09 M	0.03 M	0.04 M
SG&A	0.41 M	0.15 M	0.16 M
Leverage (total debt/total assets)	0.66	0.68	0.68
%Exports in sales	41.4%	18.5%	21.0%
%Muslim export	19.7%	0	3%
Num export destinations	22.7	5.0	7.8
Num export products (4 dg)	14.4	6.9	8.1
Num export products (6 dg)	20.3	9.7	11.4
Num new export products (6 dg)	0.25	0.14	0.17
Num new products (6 dg)*	0.42	0.26	0.30

* - new products only observable for a subsample of manufacturing firms.

soon after the boycott and identify firms that are more financially constrained because of that.

Finally, we use firm identifiers from the Firm-Integrated Database for Labor Market Research (FIDA) to match firm-level data with worker-level information from the Danish Integrated Database for Labor Market Research (IDA). IDA covers the universe of firms and workers in Denmark over 1980–2011. The data contain information about primary employment in November each year, including plant and firm identifiers, and detailed worker characteristics, allowing us to measure the skill structure of the workforce as well as pay and education of firms' managers.

The main sample uses data on all private Danish firms that were exporters during 2003–2005. The sample includes firms in agriculture, manufacturing, construction, and retail and wholesale trade, which yields a panel of about 15,000 firms. Among those, almost 1700 firms were exporting to Muslim countries and hence form our treatment group of exposed firms. Since the data is anonymized, we cannot identify the companies that we analyze. Not surprisingly, the vast majority of firms in our sample are not publicly listed, given the small size of the Danish stock market and a large number of listed firms operating in non-exporting industries.

Table 1 provides selected summary statistics. A typical firm exposed to the boycott employs 21 workers and sells over EUR 4M in total, of which almost 40% is exported. On average, 20% of their exports are sold to Muslim countries. Exposed firms are over two times larger compared to control-group firms and significantly more export-oriented. These patterns are natural given that Muslim markets are rather distant and exotic markets for Danish firms, and hence are not typically served by very small firms who export only small amounts of their production to neigh-

boring countries. While differences in employment, investment, and SG&A expenses between exposed and control firms are proportional to differences in size, the average leverage ratio is similar across groups.

3.2. Empirical strategy

Our main method to estimate firms' responses to the boycott is a difference-in-differences specification (DiD) that compares firms exposed to Muslim markets to other exporters pre- and post-boycott:

$$Y_{it} = \alpha \cdot \text{Exposed}_i \cdot \text{Post}_t + \gamma X_{it} + d_{st} + \mu_i + \epsilon_{i,t}. \quad (1)$$

The dependent variable Y_{it} measures firm outcomes such as sales or employment. The main coefficient of interest is α — the response of firms exposed to Muslim countries in the period during the boycott, ($\text{Post} = 1\{\text{year} = 2006\}$). All specifications include industry-year fixed effects d_{st} to absorb industry-level shocks and time trends. We use firm-fixed effects μ_i to capture idiosyncratic time-invariant determinants of firm outcomes and to alleviate concerns that the comparison of exposed and control firms does not capture the effect of the boycott but rather differences in pre-boycott firm characteristics.¹⁴ These characteristics may include firms' location, technology, management practices, and size, among others. We use $\ln(1 + \text{outcome})$ when analyzing flow variables such as investment, the number of new markets and the number of new products to ensure a stable sample of firm observations. We measure some outcomes (exports, sales, employment) in growth rates to allow for firm-specific trends. Accounting for trends is important in our context, because exporters to Muslim markets are generally fast-growing firms. Hence, some specifications further allow for group-specific linear trends as part of the set of time-varying firm characteristics in X . All results cluster standard errors at the industry level (for 53 distinct industries) because the shock size and firms' adjustments may interact with other firms in their industry and we want to allow for arbitrary correlation of these responses.

Table 1 reveals significant differences in size between exposed and control firms. Yet, since we include firm fixed effects and conduct our analysis in log-changes, our results are unlikely to be driven by these differences. To graphically assess any preexisting trends between these two groups of firms, we estimate an event-study regression extending Eq. (1) and subsequently plot the annual coefficient estimates capturing differences between groups, λ_t ,

$$Y_{it} = \sum_{t \in [2002, 2006], t \neq 2005} \lambda_t \cdot \text{Exposed}_i \cdot d_t + \gamma X_{it} + d_{st} + \mu_i + \epsilon_{i,t}. \quad (2)$$

where we normalize the coefficient in the year before the boycott to zero, $\lambda_{2005} = 0$. In addition, we also provide further robustness analysis around our main results, directly

¹⁴ When analyzing price responses, we perform the analysis at the firm-country-product(HS6)-year level, and include firm-country-product fixed effects. Otherwise, the specification is analogous to the firm-level specification.

controlling for firm size and other potential confounders. Finally, we rely on a matching procedure and document robust patterns within a sample matched on pre-boycott level of employment and leverage. This approach yields results similar to those obtained in the main analysis (Appendix Table A.5).

Heterogeneity. In further steps, we analyze heterogeneous responses using a triple-difference specification. In addition to dividing firms into exposure and control group, we divide them into additional groups based on heterogeneity factors. For example, we group firms into small and large based on their employment; or into low-leverage and high-leverage, based on the ratio of liabilities to assets.

Our main specification to estimate the heterogeneous response to the boycott is

$$Y_{it} = \alpha \cdot \text{Exposed}_i \cdot \text{Post}_t + \beta \cdot \text{High}_i \cdot \text{Exposed}_i \cdot \text{Post}_t + \gamma X_{it} + d_{ht} + d_{st} + \mu_i + \epsilon_{i,t}. \quad (3)$$

This model extends the DiD specification (1) by including a triple interaction of exposure, post-boycott period, and a binary variable High_i that indicates whether a firm has a high score in a given dimension of heterogeneity, for example being large (rather than small), having highly paid managers, or having high leverage. For concreteness, we will discuss the heterogeneity specification in the context of financial leverage, which turns out to be the most important dimension of heterogeneity and is analyzed more thoroughly in Section 5. The same methodology, however, is also used in the analysis of other dimensions of heterogeneity in Section 4.2.

The main coefficients of interest are α and β – the response of low-leverage firms exposed to the boycott and the differential response of high-leverage exposed firms in the period during the boycott, ($\text{Post} = 1\{\text{year} = 2006\}$). In addition to industry-year fixed effects d_{st} and firm-fixed effects μ_i , all regressions include high-leverage-by-year fixed effects, $d_{ht} = 1\{\text{year} = t\}1\{\text{high leverage} = 1\}$ to account for annual differences between low- and high-leverage firms. The identifying assumption of the interaction effect β is that absent the shock, and conditional on other controls, the difference between low- and high-leverage firms in the exposure group would have followed the same path as the difference between low- and high-leverage firms in the control group. We provide event study plots for the triple-differences specification that allow for a graphical inspection of preexisting trends between exposed and control firms by financial leverage. Specifically, we estimate

$$Y_{it} = \delta_{lte} d_{lte} + \delta_{hte} d_{hte} + d_{ht} + d_{st} + \mu_i + \epsilon_{i,t}, \quad (4)$$

where d_{lte} and d_{hte} are indicators for low- and high-leverage firms, respectively, interacted with year and exposure fixed effects (e.g., $d_{lte} = \text{Exposed}_i \cdot d_l \cdot d_t$) and we normalize $\delta_{l2005} = 0$ and $\delta_{h2005} = 0$. Finally, we provide extensive robustness by employing a matching procedure and by extending the model in (3) to include interactions of a variety of firm characteristics with exposure and year to assess whether differential adjustment by leverage in 2006 may in part reflect other differences in firm characteristics.

3.3. Heterogeneity by capital structure – identifying variation

Since a large part of the analysis will focus on capital structure, this section provides additional details about identification. We capture the differences in firms' capital structure by calculating the book leverage, defined as the ratio of total liabilities to total assets.¹⁵ Its distribution in exposure and control groups is presented in the left panel of Fig. 3. The fact that we use total liabilities and analyze exporters in a relatively bank-dependent economy explains why levels of leverage in our data are higher than typical values observed e.g., for US public firms. The two groups have similar distributions of leverage and, if anything, leverage is slightly higher for the control group. Our main measure of leverage is an indicator variable for high-leverage firms, which is defined as 1 if a firm's leverage is above the median in our main sample.¹⁶

The right panel of Fig. 3 shows the distribution of exposure to the boycott, defined as the share of exports to Muslim countries before the start of the boycott, for low- and high-leverage firms. There is no statistical difference in the degree of exposure between the two groups (The Kolmogorov–Smirnov test cannot reject the null of equal distributions with $p = 0.27$). Based on this finding and the type of variation that we observe in exposure, specifically a large number of firms with low exposure and a subset of firms with very high exposure, our main specification compares the reaction of all firms using a simple indicator of exposure. We then provide extensive robustness analysis comparing firms with high and low exposure and exploiting the full cross-sectional differences in exposure. We also use an alternative exposure measure for robustness, which is defined as the share of exports to Muslim countries in total sales (see Appendix Figure A.2).

Appendix Table A.2 provides separate descriptive statistics for low- and high-leverage firms in control and exposure groups. We report all results for the pre-boycott period 2001–2005. Similar to the size difference between exposed and control firms in the pooled analysis, there is a size difference between low- and high-leverage firms within the exposure group, as evidenced by sales, exports and employment. While firm-fixed effects will absorb any time-invariant effect of size, it is possible that size also influences the response to the boycott. While our initial analysis in Section 4.2 does not show evidence for differences in adjustments between small and large firms, Section 6.3 further alleviates this potential concern and shows that the effect of leverage does not capture the differential size of firms or breadth of their product portfolio. In particular, the striking differences by leverage also remain after matching high- to low-leverage exposed firms based on firm size.

¹⁵ To avoid any contamination of our measure by the early stage of the boycott and expectations of the boycott by firms or lenders in late 2005, we use balance sheet information at the end of the 2004 fiscal year to construct our time-invariant leverage measure used throughout the analysis.

¹⁶ While we use the country-wide median in the baseline specification, Appendix Table A.11 demonstrates that the results are robust to defining the median by industry and to using continuous measures of leverage, among other alternatives.

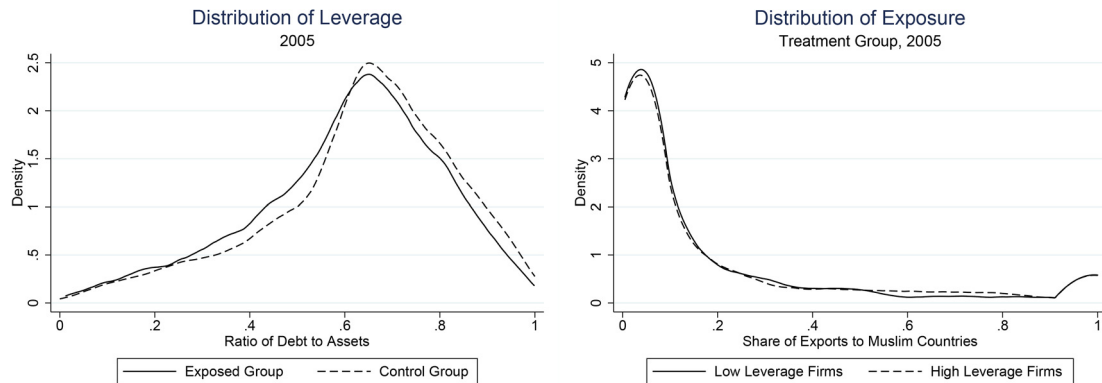


Fig. 3. Sample balance by exposure and leverage. The left figure shows the distribution of total leverage (defined as the ratio of total liabilities to total assets), separately for firms exposed to the boycott (exposed group, firms with at least 0.5% of exports sold to Muslim countries) and those not exposed (control group). The distributions are smoothed using Epanechnikov kernel with bandwidth 0.035. Observations with leverage above 1 are censored. The right figure shows the share of exports sold to Muslim countries for low- and high-leverage firms. High leverage is defined as above-median share of total liabilities in total assets and the distributions are smoothed using Epanechnikov kernel with bandwidth 0.04.

Analyzing how debt affects the response to the boycott relates to a traditional problem of the corporate finance literature: distinguishing between economic and financial distress (Asquith et al. 1994; Andrade and Kaplan 1998). To tackle this problem, we use a difference-in-differences strategy and explicitly control for the consequences of economic distress. Our approach is conceptually similar to that taken by Giroud and Mueller (2017), who compare the reaction of low- and high-leverage firms to local demand shocks driven by the financial crisis, and to Kini et al. (2017), who compare how two negative industry-level shocks influence firms' probability of product recall depending on their initial leverage. We compare the differential response of firms with similar exposure to the boycott, but with different leverage before the crisis. We attribute any differences between low- and high-leverage firms to the impact of different capital structure.

Another traditional empirical challenge is endogeneity of leverage choice, that may be driven by reverse causality or omitted variables. Our research design is well-suited to rule out reverse causality concerns. The unexpected nature of our shock did not allow firms to adjust their capital structure in anticipation of the boycott. Concerns about omitted variables are alleviated by our triple-difference design. Because we explicitly control for the difference between low- and high-leverage firms not subject to the shock, our results cannot be attributed to an omitted factor which would lead to diverging outcomes of low- and high-leverage firms even in the absence of the boycott. Some omitted factor may, however, influence firms' reaction to the boycott. To alleviate this concern, we employ two strategies. First, we rely on information on loan maturity to identify firms for whom a large share of debt happens to mature soon after the boycott. This quasi-experimental approach to isolate firms subject to financial constraints allows us to confirm our main results relying on an alternative source of variation. Second, we directly control for the role of several main confounding factors in the response to the boycott by exploiting our rich data on

managers' pay and education, production technology and business model, and firms' past performance. We provide more details about these approaches in Section 6.3.

4. Response to the boycott

4.1. Average response to the boycott

The top panel of Fig. 4 shows the annual coefficients for exports to Muslim- and non-Muslim destinations for firms in the exposure group. After the boycott, there is a significant decline in the exports to Muslim markets, consistent with the aggregate drop in exports to Muslim countries presented in Fig. 1. That drop is accompanied by an increase in exports to non-Muslim markets.

Columns 1 and 2 of Table 2 complement the graphical evidence and report the average effect of the boycott on exports to Muslim countries analyzing firms in the exposure group only. Column 1 documents a 51 log points/40% decline in affected firms' exports to Muslim markets. That number, however, underestimates the impact of the boycott, because it does not account for the fact that 26.2% of exposed firms stopped exporting to Muslim countries all together and hence do not contribute to the estimated effect on log exports in column 1. To combine both the intensive and extensive margin effect, we compute arc-percent changes of exports. Arc-percent changes are defined as the change between two periods divided by the average value of the variable in these two periods, $\text{arc-percent change}(x)_t = \frac{x_t - x_{t-1}}{(x_t + x_{t-1})/2}$, and hence can take a value between -2 and 2 . Column 2 demonstrates a -1.27 drop of exports to Muslim countries using arc-percent changes, and this large coefficient further emphasizes the role of market exit.

Taken together, columns 1 and 2 demonstrate a dramatic decline of export activity to Muslim markets, implying an average decrease in the range of 56–78% of ex-

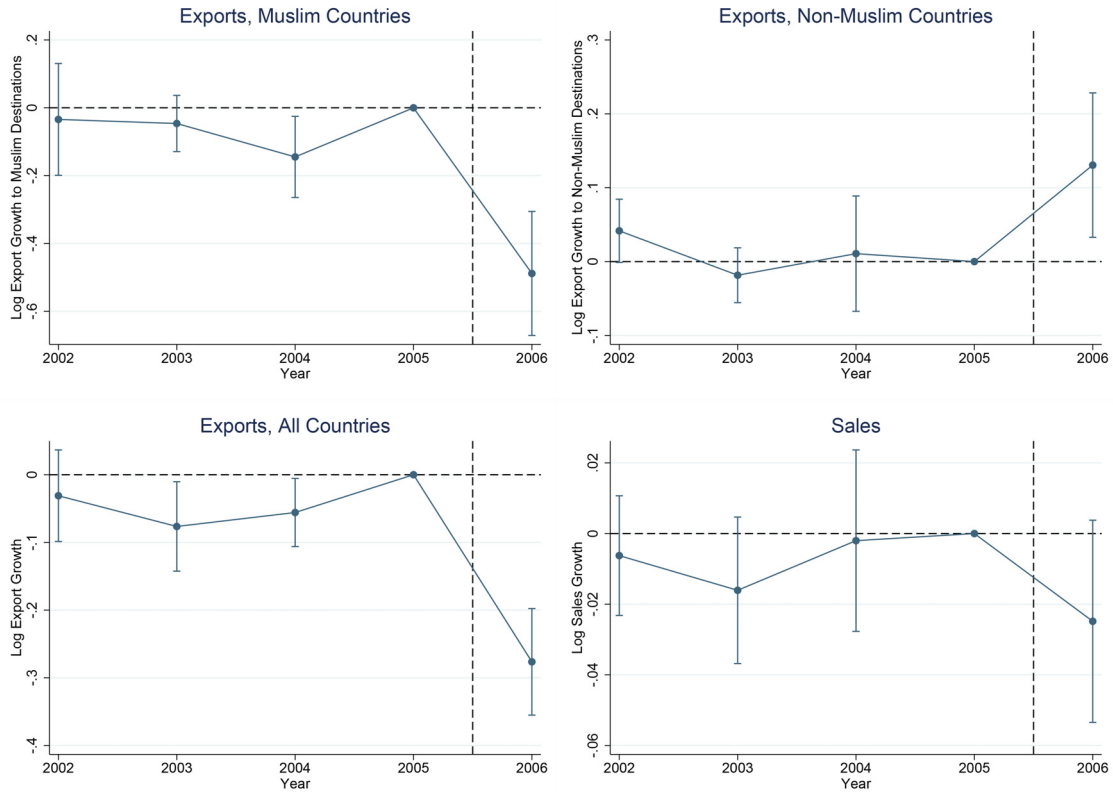


Fig. 4. Exports and sales response to the boycott. The figures plot λ coefficients from regression (2) and 95 percent confidence intervals, which are clustered at the industry level. The left and right graphs in the top row show results for exports to Muslim and Non-Muslim markets, while the bottom row presents coefficients for total exports and total sales regressions.

Table 2

Impact of the boycott.

Columns 1–3 contain only firms exposed to the boycott and estimate Eq. (2) with firm fixed effects and year-fixed effects. Columns 4–9 include the full sample of treatment and control firms and estimate the main pooled difference-in-differences model described in Eq. (1), including firm fixed effects and industry-year fixed effects. The main independent variable is the interaction of exposure to the boycott (exposed) and post-boycott period (year 2006). Dependent variables in columns 1, 3–8 are log-differences in exports to Muslim countries, exports to non-Muslim countries, total exports, number of country-product pairs a firm exports to, product price, total sales, and total wage bill. In column 2, the dependent variable is the arc-percent change of exports, defined as $(Export_t - Export_{t-1}) / [(Export_t + Export_{t-1}) / 2]$. Column 6 analyzes log changes in unit prices (price per kilogram) at the firm-country-product (HS6)-year level; we include firm-country-product (HS6) fixed effects. Column 9 analyzes the return on assets, defined as net income to total assets. Column 10 is a cross-sectional regression of an indicator for firm death in 2006; we include industry fixed effects and lagged firm size controls. The last row reports the sample average of the dependent variables (first-differences). For columns 1–9, we report the average over 2002–2005 before the boycott. For column 10 we report the share of firms that exited in 2006. Sample size in columns 1–3 reflects only firms exposed to Muslim markets; columns 1 and 3 have lower sample size because logarithms are not defined for firms with zero flows. In columns 4–5, 7–9 all firms are included, but sample size varies because export flows are zero in some years for some firms, and because of missing data. All regressions cluster standard errors at the industry level (53 industries), *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1) $\Delta \ln(\text{Exp Muslim})$	(2) Arc-Change Exp Muslim	(3) $\Delta \ln(\text{Exp Other})$	(4) $\Delta \ln(\text{Exp Total})$	(5) $\Delta \ln(\text{Num Ctry-Prod Pairs})$	(6) $\Delta \ln(\text{Unit Prices})$	(7) $\Delta \ln(\text{Sales})$	(8) $\Delta \ln(\text{Wages (Total)})$	(9) ROA	(10) Firm Exit 2006
Year = 2006	-0.5110*** (0.089)	-1.2720*** (0.090)	0.1471*** (0.049)							
Exposed X 2006				-0.2687*** (0.042)	-0.1288*** (0.036)	-0.0226 (0.025)	-0.0294*** (0.011)	-0.0324** (0.012)	0.0026 (0.006)	0.0007 (0.003)
Obs	4792	6472	6955	44,698	44,698	631,988	61,376	59,346	61,375	12,230
R-squared	0.024	0.151	0.007	0.013	0.014	0.183	0.003	0.010	0.007	0.009
Firms	1424	1675	1628	13,019	13,019	8368	14,900	14,522	14,900	12,230
Sample Avg	0.163	0.356	-0.009	0.007	0.061	-0.0235	0.046	0.038	0.076	0.020

ports.¹⁷ To compare these changes to total revenue, we note that the share of exports in total sales was 41.4% for the exposed firms in the pre-period (Table 1) and almost 20% of exports were directed to Muslim countries. In addition, exports to Muslim countries were growing before the boycott at a rate of 16.3%, while exports to other markets were stable. Hence, in 2006 export to Muslim countries was expected to constitute 9.5% of total sales of exposed firms. Combining the estimated 56–78% export decrease with the average importance of these markets for exposed firms, the boycott has likely decreased affected firms' sales by around 6%, compared to a counterfactual no-boycott scenario. Yet, this number does not take into account any adjustments that affected firms may take, including increasing sales in other markets, which we will now investigate in more detail.

The lower panel of Fig. 4 plots λ coefficients from regression (2) for the difference between firms in the exposure and control group for total exports and sales over time. The graphical evidence shows a large and statistically significant decline in total exports following the boycott, and a smaller decline in total sales that is only significant at 10% level. Sales differences across groups before the boycott are economically small and statistically insignificant, whereas total exports growth was accelerating for exposed firms before the boycott, compared to control firms. In light of this trend, the drastic decline in total exports in 2006 is even more striking. Estimating pooled difference-in-differences model (1), column 4 of Table 2 shows that firms exposed to Muslim markets export ~24% less in 2006, compared to non-exposed firms. This effect is consistent with the large decline of exports to Muslim markets documented in columns 1 and 2, and is further confirmed by a significant decline in the number of product-country pairs served (column 5, ~12% decline).

Unit prices at the HS-6 product level did not change differentially among exposed firms after the boycott (column 6). The specification in column 6 includes firm-country-product fixed effects, and hence isolates price changes for products that a firm sold in a particular market both before and after the boycott. Appendix Table A.3 provides additional results on the evolution of prices. Across various specification, we find a robust lack of pricing adjustments for existing products. Analyzing prices of products introduced or discontinued after the boycott, we find that they were generally more expensive compared to products sold both before and after the boycott. Yet, higher prices for new products are not specific to firms affected by the boycott and likely reflect different product quality or other

factors driving price differences across the product life cycle.¹⁸

In total, the pooled estimates in columns 7 and 8 of Table 2 suggest that the boycott led to a decline in sales and employment, measured by the total wage bill,¹⁹ by around 3%. Profitability, measured by the return on assets, does not significantly change after the boycott (column 9), consistent with the fact that affected firms mitigate the impact on short-term profits by recouping lost revenues or cutting costs. Short-term profits, however, are unlikely to be a good measure of the optimality of firm's response. While both expanding revenues and cutting costs may have a similar impact on short-term profitability, these adjustments may have different implications for long-term value creation. In the long term, redirecting sales and investing in market share might be profitable, but might also constitute overinvestment. The shock is also too small to significantly affect the likelihood of firms' bankruptcy. A cross-sectional specification for 2005 does not show a significant increase in the likelihood of firm exit in 2006, and the point estimate is small compared to the 2% probability of exit in 2006 (column 10). All results are similar when using a matching procedure to narrow down the control group (Appendix Table A.5, Panel A).

As discussed earlier, observed magnitudes of the decline in exports to Muslim countries suggest a total sales drop of around 6%. Yet, we only observe half of that decline, suggesting that affected firms increase domestic sales (which also include sales to other European markets up to a threshold) to make up for sales lost in Muslim markets. While the data do not allow us to independently measure domestic sales, a similar logic applies to exports to non-Muslim markets, which we can analyze using our export data. Column 3 of Table 2 demonstrates the estimates for log-changes of exports to non-Muslim markets. The point estimate of the effect of the boycott is large, suggesting an almost 16% increase in sales to other markets on average. The 90% confidence interval ranges from 5 to 25% increase, which suggests that the effects on sales to other markets may be more nuanced and display significant heterogeneity across firms. In the next section, we analyze the heterogeneity of the response in more detail.

4.2. Heterogeneity analysis

Table 2 shows that the boycott caused a significant decline in sales and employment for an average exposed firm, but it is likely that not all firms were affected and re-

¹⁷ Column 1 suggests that firms that continued to export to some Muslim markets see a 40% decline in sales, which together with 26% firms losing their entire exports suggests an average drop of 56%. Column 2 implies a 78% exports drop on average, if we normalize exports in 2005 to 1, substitute it as x_{t-1} in arc-percent change definition, and solve for x_t given the -1.27 change in column 2 ($-1.27 = \frac{-0.78}{(1+0.22)/2}$). Alternatively, note that the arc-percent change measure would be -2 if all firms lost all their exports to Muslim markets, and hence for large negative changes dividing by 2 may provide a reasonable approximation of the change in percentage terms (-63.5%).

¹⁸ It should be noted that our price measures are unit prices based on the shipment value and quantities from customs declarations. Well-known limitations of these measures include potential measurement error in quantity data and changes in the quality mix of heterogeneous goods within HS product category. In addition to these caveats, customs declarations may fail to include some discounts or special contract provisions (such as financing or service conditions) that Danish firms may offer to their buyers after the boycott.

¹⁹ The total wage bill is useful to precisely measure the size of labor inputs, but changes in total wage bill may reflect both the changing size of the workforce and changes in average wage. In our context, the latter does not seem to play a significant role. We analyze wage responses and other measures of employment in more details in the Appendix Table A.9.

sponded in the same way. In this section, we analyze heterogeneous adjustment to the boycott along four key dimensions: firm size, management quality, product variety, and financial leverage. Adaptation may differ by firm size because larger firms may have more resources and margins of adjustments available, and hence can react to the boycott differently; at the same time, small firms may have better relations with their customers and may be less affected by the boycott. Better managers, where quality is proxied with management salary, may be more successful in alleviating the effects of the boycott and adjusting to it. Firms with larger product portfolio may be able to redirect their sales more easily. Financial leverage may determine firms' ability to obtain capital needed to finance adjustments to the shock.

To illustrate differences in adjustment across firm types, we define two groups (high and low) for each firm characteristic of interest and employ a triple-difference specification, see Eq. (3). Table 3 reports the results. Panel A compares small and large firms based on firms' employment. While the coefficients on the triple interaction term suggest that large firms do slightly better in terms of total sales and employment, despite losing a larger share of their exports to Muslim markets, none of these coefficients are statistically significant, and their economic magnitudes are limited. A similar picture arises in Panel B, which explores heterogeneity with respect to manager pay. Again, while the point estimates suggest that firms with higher-paid managers do slightly better in terms of total sales and employment, these differences are statistically insignificant and have modest magnitudes.

Panel C compares firms with various sizes of their product-market portfolio, categorizing firms as those with below- and above 100 product-market pairs in the pre-boycott period. Despite being hit by the same shock (column 1), firms with larger product-market portfolio do better in terms of total exports (column 3). Interestingly, their relative increase in exports to non-Muslim countries is slightly smaller, but that is mainly because their base level of non-Muslim exports is significantly higher. Overall, firms that are already serving many markets before the boycott further increase the number of product-country pairs in their export portfolio, despite losing some sales in Muslim markets (the sum of coefficients in column 4 is positive). Observed differences are consistent with the importance of market entry costs: with higher existing presence in export markets, it may be easier and cheaper to find additional demand when the boycott creates excess production capacity. Yet, this advantage is not large enough to avoid significant declines in total sales and employment – firms with higher market presence perform slightly better along those margins, but the differences are not significant.

Panel D compares firms with above- and below-median financial leverage. This dimension of heterogeneity reveals large and important differences between the two subsets of exporters. While both low- and high-leverage firms are hit by the same shock (column 1), low leverage firms are more successful at increasing exports to other markets (column 2), and hence total exports decline more for high-leverage firms (column 3). A similar, yet even clearer pattern is visible for the number

of product-country pairs (column 4). While both groups are boycotted by the same set of Muslim countries, low-leverage firms respond by introducing new products into new non-Muslim destinations, and as a result, their total number of product country-pairs remains unchanged. High-leverage firms, on the other hand, do lose a significant number of product-country pairs. These differences in export behavior, and possibly analogous differences in domestic sales, translate to different patterns of total sales and employment. While both sales and employment decline for high-leverage firms, there is no significant impact on low-leverage firms (columns 6, 7). Notably, the significant difference between high and low leverage firms appears in market entry, not in price adjustments. Similar to other dimensions of firm heterogeneity, we find no evidence of differential price adjustments by leverage (column 5), consistent also with the lack of a significant price response of an average firm. Analyzing firm profitability (column 8), we find that the effect on ROA is lower for high-leverage firms and the difference is significant at a 10% level. Yet, as discussed in the previous section, short-term profitability is unlikely to be a good measure of the optimality of firms' response.

The heterogeneity of the boycott effect by leverage is striking: while the shock is equally large for low- and high-leverage firms, their response and the final impact of the boycott are very different. Low-leverage firms are able to fully make up for lost demand and weather the shock without reducing employment. High-leverage firms, however, are significantly adversely affected. To shed more light on these differences, Section 5 analyzes the heterogeneous response by leverage in more detail.

4.3. Robustness

We revisit our results on the effects of the boycott in Appendix Table A.4, which analyzes several alternative specifications. Specifically, we first add controls for time-varying measures of firm size and product portfolio in panels A and B. To avoid mechanical bias in the estimates, we lag these control variables by two periods. Across these extensions, we find robust and precisely estimated negative effects on total exports, sales, and employment. Our point estimates for the overall effects on sales range from 2 to 4%, placing our main result from Table 2 around the midpoint of this interval.

In panel C, we replace the binary indicator for boycott exposure with a continuous measure of the share of exports to Muslim countries in 2005. Panel D uses an alternative measure of exposure, i.e., the share of exports to Muslim countries in total sales (as opposed to total exports). Since these two measures of exposure have different distributions among exposed firms, we simplify the interpretation of these results by first standardizing these measures by dividing by their standard deviation. As expected, the results show substantially larger losses in exports to Muslim countries for firms with higher previous share of business in these markets. The effects of one standard deviation higher exposure is similar in magnitude to the average effects in Table 2, which confirms that the choice of exposure measure is not driving our results.

Table 3

Heterogeneity of the impact of the boycott.

All regressions estimate the triple difference specification (3), but the dimension of heterogeneity differs. The main independent variables are interactions of the firm being exposed to the boycott and post-boycott indicator; and triple interaction of exposure, post-boycott, and scoring high on selected dimension of heterogeneity: size, based on employment above/below 100 (panel A); management pay above/below median (panel B); number of product-country pairs above/below 100 (panel C); and book leverage above/below median (panel D). Dependent variables are log-changes in exports to Muslim countries, exports to Non-Muslim countries, total exports, number of 6-digit product-country pairs served, unit prices (price per kilogram), total sales, and total wage bill, as well as the return on assets. All regressions are at the firm-year level except column (5), where observations are at the firm-product (HS6)-country-year level and we include firm-HS6-product-country fixed effects. Sample size in columns 1–2 reflects only firms exposed to Muslim markets. In columns 3–4, 6–8 all firms are included, but sample size varies because export flows are zero in some years for some firms, and because of missing data. In particular, data on management pay is only available for a subsample, resulting in significantly smaller sample sizes in Panel B. All regressions cluster standard errors at the industry level (53 industries), *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Exposed firms		All firms					
	$\Delta \ln(\text{Exp Muslim})$	$\Delta \ln(\text{Exp Non-Muslim})$	$\Delta \ln(\text{Exp Total})$	$\Delta \ln(\# \text{Ctry-Prod Pairs})$	$\Delta \ln(\text{Unit Prices})$	$\Delta \ln(\text{Sales})$	$\Delta \ln(\text{Wages})$	ROA
Panel A: Employment > 100 FTE								
Exposed	-0.4990***	0.1335**	-0.2565***	-0.0762***	-0.0051	-0.0190	-0.0178	0.0038
X 2006	(0.100)	(0.054)	(0.044)	(0.027)	(0.021)	(0.012)	(0.013)	(0.007)
Exposed	-0.0478	0.0614	0.0864	-0.0093	-0.0101	0.0241	0.0089	-0.0026
X High	(0.131)	(0.1044)	(0.056)	(0.050)	(0.070)	(0.021)	(0.024)	(0.010)
X 2006								
Obs	4744	6894	42,648	42,648	615,909	58,693	56,861	58,693
Panel B: Management Pay > Median								
Exposed	-0.6772***	0.0098	-0.3036***	-0.0876***	0.0119	-0.0607***	-0.0682***	0.0018
X 2006	(0.174)	(0.046)	(0.057)	(0.023)	(0.035)	(0.016)	(0.010)	(0.008)
Exposed	0.0991	-0.0620	0.0085	-0.0133	-0.0113	0.0277	0.0105	-0.0031
X High	(0.112)	(0.074)	(0.083)	(0.061)	(0.077)	(0.025)	(0.014)	(0.008)
X 2006								
Obs	3525	4818	23,954	23,954	541,443	29,706	29,527	29,706
Panel C: Market Access: > 100 Product-Country Pairs								
Exposed	-0.5095***	0.1527**	-0.2753***	-0.0645**	0.0122	-0.0354***	-0.0339**	0.0017
X 2006	(0.108)	(0.058)	(0.042)	(0.029)	(0.019)	(0.012)	(0.012)	(0.007)
Exposed	-0.0055	-0.0270	0.2358***	0.2126***	-0.0493	0.0163	0.0086	0.0045
X High	(0.088)	(0.052)	(0.046)	(0.045)	(0.048)	(0.018)	(0.012)	(0.009)
X 2006								
Obs	4792	6955	44,690	44,690	631,988	61,362	59,334	61,361
Panel D: Book Leverage > Median								
Exposed	-0.5372***	0.2502***	-0.2053***	-0.0169	-0.0423	0.0028	-0.0133	0.0098
X 2006	(0.107)	(0.064)	(0.052)	(0.036)	(0.050)	(0.010)	(0.016)	(0.006)
Exposed	0.0614	-0.2328***	-0.1360**	-0.1004***	0.0535	-0.0656***	-0.0377**	-0.0145*
X High	(0.094)	(0.055)	(0.060)	(0.035)	(0.068)	(0.023)	(0.016)	(0.008)
X 2006-								
Obs	4771	6925	43,706	43,706	629,643	60,138	58,240	60,138

These larger shocks in Muslim markets translate to a larger decline in total exports, but the overall effects on sales and employment are less closely related to continuous exposure measures. This evidence lends support to the key role that firm heterogeneity plays in adapting to this shock.

Panel E focuses on firms with stable preexisting business in Muslim markets. Specifically, we focus on the subsample of firms with exports to Muslim countries in both 2004 and 2005. We further distinguish firms within this group by their level of exposure, using a threshold of 10% of exports to these markets in both years to define high exposure. The results suggest that the proportional decline in exports to Muslim destinations was similar for firms with low and high exposure. Yet, these shocks account for a different share of total exports and thus lead to larger decline in total exports for highly exposed firms. Moreover, these

shocks then translate to a larger decline in sales and wage bill, even though the effects for this relatively small set of firms with high exposure are sometimes less precisely estimated.

5. Heterogeneity by capital structure

Table 3 suggests that low-leverage firms were able to increase their exports to other markets and adjust to the boycott without losing much of their business, but high-leverage firms were not. In this section, we characterize these differences in more detail and analyze the differences in market entry patterns by leverage. Because we define a market as a country-product pair, we further explore both the country and product dimension of the response. Finally, to support the financial constraints interpretation

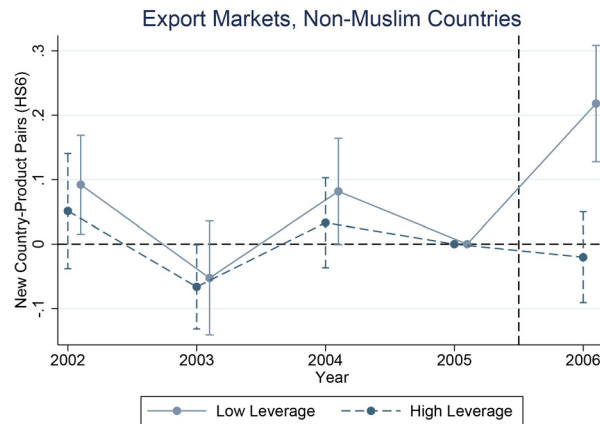


Fig. 5. Market entry response. The dependent variable is the change in the log of 1+ number of new product-country pairs for non-Muslim countries. The bars in the figure show coefficients and 95% confidence intervals from a regression analogous to the main specification in which high-leverage $\times$ exposed $\times$ pre-boycott term was split into several terms for each year separately (normalizing the year 2005). Coefficients for low- (high-) leverage firms are for firms exposed to the boycott with leverage below (above) the median.

of our results, we provide additional evidence on investment, SG&A, borrowing, and employment responses.

5.1. Market entry

Panel D of Table 3 demonstrates that low- and high-leverage firms reduce exports to Muslim countries to a similar extent (column 1), yet low-leverage firms do significantly better in terms of total exports and total sales (columns 3 and 5). These results suggest that low-leverage firms are more successful in redirecting their sales to other export markets and increasing domestic sales. To shed more light on this process, we turn to analyzing product-level data.

Figure 5 graphically demonstrates the most important result of this section, presenting coefficients from an extended regression with annual coefficients for low- and high-leverage firms in the exposure group; see Eq. (4). The solid line shows that low-leverage firms respond to the boycott by increasing the number of non-Muslim product-country pairs in which they are present. In contrast, the response for high-leverage firms (dashed line) is close to zero and insignificant.

Table 4 provides a more detailed analysis of firms' entry into new markets. In columns 1 and 2 we analyze the number of product-country pairs in non-Muslim markets, for which we start observing non-zero export flows for a given firm. Column 1 defines product category based on the 6-digit classification from the harmonized system (HS), while column 2 uses the 4-digit classification. An example of 4-digit categories include "Coffee" and "Tea," while "Black Tea" or "Green Tea" are examples of 6-digit products. Columns 3 and 4 look at the total number of markets (product-country pairs) served by the firm, based on 6- and 4-digit HS codes. Columns 5–7 further explore the two dimensions of a "market": number of countries in which the firm is present (column 5), and number of products that it sells (columns 6 and 7).

Our results confirm that low-leverage firms increase their entry into new non-Muslim markets (columns 1 and 2), which allows them to keep the total number of product-country pairs unchanged (columns 3 and 4): due to the boycott, the total number of countries they serve decreases (column 5), but they stabilize the number of markets they serve in part with an increase in the number of exported products (columns 6 and 7). What kind of adjustment do these various changes reflect? One example is the following situation (stylized for confidentiality reasons): a Danish food producer was selling product A to Saudi Arabia and Qatar before the boycott (1 product, 2 countries). After the boycott, access to these two markets was lost, and sales of product A were partially redirected to Russia. However, presumably because the demand for product A in Russia was limited, only half of the previous amount could be sold there. Instead, however, there might have been demand for technologically-related product B. Therefore, after the boycott the exporter was selling 2 products to 1 country: the number of destinations decreased, the number of total product-country pairs remained unchanged, but the number of products went up.

The market entry response is muted for high-leverage firms. They do not significantly expand to non-Muslim markets (total effect in columns 1 and 2 is close to zero), and as a result they see a significant decline in the number of total product-country pairs (columns 3 and 4) and a larger decline in the number of export destinations (column 5). Their total number of exported products remains unchanged (columns 6 and 7), suggesting that products boycotted by Muslim countries were also sold elsewhere.

5.2. Country dimension of the market entry response

What are common characteristics of countries where new market entry is most pronounced? Fig. 6 presents market entry responses of low- and high-leverage firms distinguishing new exports into different groups of destination countries defined by region, size, geographic and

Table 4

Redirecting sales: new export markets and products.

All regressions are based on Eq. (3) and include firm fixed effects, industry-year fixed effects, and binary variables for each year interacted with indicators for high leverage. The main independent variable is the triple interaction of being exposed to the boycott (exposed), having high leverage and post-boycott period (year 2006). Dependent variables are log-changes of 1 + the number of new non-Muslim markets, number of total product-country pairs (defined as non-zero flows in 6-, 4-digits HS product category and a given country), number of export destinations, and number of exported products. The bottom row presents mean levels of dependent variables in the pre-boycott period. “High - Total Effect” presents the sum of baseline coefficient for low-leverage firms and differential effect for high-leverage firms and the *p*-value row presents the corresponding *p*-value from the *F*-test for significance of high-leverage firms’ total response. In all regressions, standard errors are clustered at the industry level, *** *p* < 0.01, ** *p* < 0.05, * *p* < 0.1.

	(1) Δln(1+New Prod-Ctry Pairs, Non-Muslim)	(2) Δln(1+New Prod-Ctry Pairs, Non-Muslim)	(3) Δln(Num Product-Ctry Pairs)	(4) Δln(Num Product-Ctry Pairs)	(5) Δln(Export Destinations)	(6) Δln(Num Export Products)	(7) Δln(Num Export Products)
	6 digits	4 digits	6 digits	4 digits		6 digits	4 digits
Exposed X 2006	0.1896*** (0.034)	0.1837*** (0.039)	−0.0169 (0.036)	−0.0309 (0.032)	−0.1277*** (0.024)	0.0977*** (0.031)	0.1105*** (0.027)
Exposed X High X 2006	−0.2137*** (0.046)	−0.2082*** (0.041)	−0.1004*** (0.035)	−0.0978*** (0.032)	−0.0720*** (0.017)	−0.1116*** (0.038)	−0.1152*** (0.036)
High - Total Effect (<i>P</i> -val)	−0.0241 0.370	−0.0245 0.317	−0.117*** <0.001	−0.129*** <0.001	−0.200*** <0.001	−0.0139 0.473	−0.00466 0.813
Observations	43,706	43,706	43,706	43,706	43,706	43,706	43,706
R-squared	0.035	0.030	0.044	0.043	0.033	0.036	0.033
Firms	12,255	12,255	12,255	12,255	12,255	12,255	12,255
Sample Avg	10.91	7.576	32.15	24.11	7.815	11.44	8.114

linguistic distance to Denmark, and by existing export relationships between these countries and Denmark or a specific Danish exporter. Each pair of bars presents coefficient estimates of the response of low- and high-leverage firms from separate regressions that analyze new market entry in different groups of countries with a specific common characteristic, respectively. Since the market is defined as product-country pair for which the firm has positive sales, the coefficients represent both entering new countries and introducing new products into countries where the firm was selling other products before.

Panel A of Fig. 6 analyzes the role of existing export relationships. The top-left panel demonstrates that the market entry response of low-leverage firms is significantly larger for countries with above-average level and past growth in exports of all Danish firms before the boycott. This result suggests that the response of low-leverage firms targets markets where other Danish firms are already present, and in which Danish firms were growing faster in the past. This may be because firms have better information about these markets, it is cheaper and easier to establish trade connections there, or because these markets reflect parts of the global economy where demand for Danish products is the highest. The top-right panel suggests that post-boycott market entry of low-leverage firms is also slightly stronger for markets where a given firm itself sold more in the past, although the difference between above- and below-average group is modest. Interestingly, however, the opposite pattern is visible for firm-level growth patterns: low-leverage firms increase market entry twice as much in countries in which their past growth was below average, although the confidence intervals of the two coefficients overlap. This suggests that firms target places with untapped potential and may try to enter markets in which they have not been growing as fast as they possibly could. Taken together, these results illustrate

an interesting trade-off: familiar markets may be easier to enter, but past expansions may limit additional growth potential in a given country.

It is interesting to note that when we zoom into specific types of markets, we do find also a significant market entry response by high-leverage firms. In particular, high-leverage firms significantly increase their entry into markets in which their existing export level or export growth were above average. In all cases, however, the magnitude of their expansion is significantly smaller than that of low-leverage firms. These results lend an indirect support to our financial constraints interpretation of the main results: high-leverage firms presumably find it optimal to expand, as they do so in selected markets, but they are only able to do so for countries where they are already present, consistent with lower market entry costs there.

Panel B analyzes the role of geography, language, and market size. While we find no significant effects in any of the country groups for high-leverage firms, we document significant effects for low-leverage firms in a large variety of market types. Geographically, the expansion of low-leverage firms is quite broad, with three out of four world regions – except Africa – showing a similar, significant response. Along other margins, the market entry response is also broad-based, with all coefficients being significant for countries both above and below average in terms of GDP per capita, population size, distance to Denmark, and share of population in Denmark and in the destination country that speaks a common language. The magnitudes of coefficients suggest that firms respond more strongly in countries that are more lucrative and easier to enter: richer, larger, and more similar in terms of language. These patterns are also true for high-leverage firms, although their response is always insignificant. Interestingly, firms’ responses are very similar for countries that are close and far from Denmark. This, however, hides some heterogene-

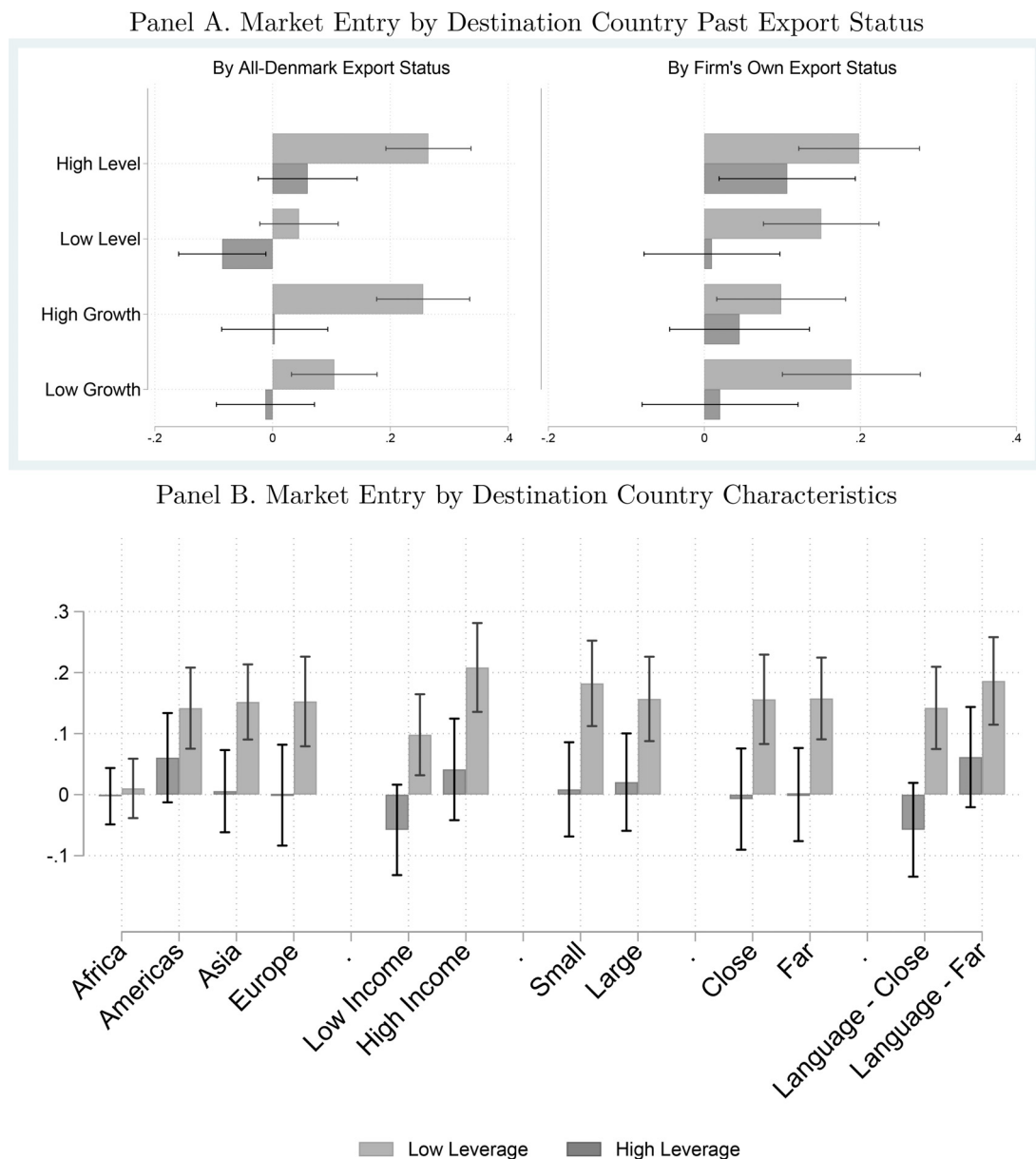


Fig. 6. New market entry by country characteristics. Both panels show coefficients and 95% confidence intervals from regressing $\log(1 + \text{number of new non-Muslim product-country pairs in a given group of countries})$ on the indicator for being exposed to the boycott, and being exposed and having high-leverage (main specification). Groups of countries are defined as follows: in the top panel, markets are divided by past export pattern to the destination country: above and below average export levels and growth rates of exports from Denmark (left) and by the analyzed firm (right). In the bottom panel, groups of countries are defined based on geography (4 regions) and on country being above or below average in terms of GDP per capita, population size, distance to Denmark, and share of population speaking common language as Denmark (the measure accounts not just for Danish speakers, but also for the fact that Danish people sometimes speak English and other languages).

ity: additional analysis reveals that the response is strong for the closest regions, such as Scandinavia, but is also sizable for large but distant markets such as China or United States, making the average effect similar independent of the distance.

Overall, column 5 of Table 4 demonstrates that the boycott leads to a significant decline in the number of countries in which the firm is present, which is of course caused by exit from Muslim markets. For high-leverage firms, this boycott leads to a 20% decline in the number

of countries served. Low-leverage firms, however, are able to alleviate that decline by entering new countries, and as a result the number of their export destinations decreases only by ~13%.

5.3. Product dimension of the market entry response

The second dimension of the market entry response are the products that are being sold in new markets. In Table 5 we explore how different products that the firm has sold

Table 5

Market entry: product response.

All regressions are based on Eq. (3) and include firm fixed effects, industry-year fixed effects, and binary variables for each year interacted with indicators for high leverage. The main independent variable is the triple interaction of being exposed to the boycott (exposed), having high leverage and post-boycott period (year 2006). Dependent variables are a binary indicator for trade redirection: firm's export of a given product declining in Muslim countries and increasing in non-Muslim countries (column 1); log-changes of 1 + number of new product-country pairs (defined as non-zero flows in 6-digit HS product category and a given country), for non-Muslim markets that fall within above/below average group defined based on the share of new entrants among all sellers (columns 2 and 3); export levels (columns 4 and 5) or export growth (columns 6 and 7) of a given firm. The bottom row presents mean levels of dependent variables in the pre-boycott period. "High - Total Effect" presents the sum of baseline coefficient for low-leverage firms and differential effect for high-leverage firms and the *p*-value row presents the corresponding *p*-value from the *F*-test for significance of high-leverage firms' total response. In all regressions, standard errors are clustered at the industry level, *** *p* < 0.01, ** *p* < 0.05, * *p* < 0.1.

	(1) Trade Redirection	(2) Above	(3) Below	(4) Above	(5) Below	(6) Above	(7) Below
		Δln(1 + New Prod-Ctry Pairs, Non-Muslim, 6 digits)					
		% of Entry		Firm Avg Level: HS-2		Firm Avg Growth: HS-2	
Exposed X 2006	0.0145*** (0.002)	0.2449*** (0.037)	0.1712*** (0.036)	0.1293*** (0.040)	0.1267*** (0.043)	0.0896* (0.050)	0.1720*** (0.058)
Exposed X High X 2006	−0.0063*** (0.002)	−0.1304** (0.053)	−0.2047*** (0.054)	−0.1497** (0.063)	−0.1362** (0.064)	−0.1078 (0.074)	−0.1547* (0.086)
High - Total Effect (<i>P</i> -val)	0.00829*** 0.001	0.114*** 0.003	−0.0334 0.419	−0.0204 0.681	−0.00954 0.842	−0.0182 0.739	0.0173 0.785
Observations	105,142	43,706	43,706	43,706	43,706	43,706	43,706
<i>R</i> -squared	0.309	0.028	0.041	0.017	0.018	0.019	0.013
Firms	1575	12,255	12,255	12,255	12,255	12,255	12,255
Sample Avg	0.0245	2.836	10.24	6.190	5.128	5.450	5.868

Table 6

Product portfolio adjustments.

All regressions are based on Eq. (3) and include firm fixed effects, industry-year fixed effects, and binary variables for each year interacted with indicators for high leverage. The main independent variable is the triple interaction of exposure to the boycott (exposed), having high leverage, and post-boycott period (year 2006). Dependent variables are the log-change in the 1 + number of new products sold at HS-6 and HS-4 level; an indicators for any new products sold, number of new product sold, and number of new product sold that are also newly exported (all at HS-6 level). The bottom row presents the mean of dependent variables in the pre-boycott period. "High - Total Effect" presents the sum of baseline coefficient for low-leverage firms and differential effect for high-leverage firms and the *p*-value row presents the corresponding *p*-value from the *F*-test for significance of high-leverage firms' total response. In all regressions, standard errors are clustered at the industry level, *** *p* < 0.01, ** *p* < 0.05, * *p* < 0.1.

	(1) Δln(1 + # New Products) (6 digit)	(2) Any New Product	(3) Δ# New Products	(4) Δ# New Export Products
Exposed X 2006	0.1028** (0.049)	0.0814* (0.040)	0.0305* (0.015)	0.5448** (0.238)
Exposed X High X 2006	−0.1095** (0.041)	−0.0881** (0.042)	−0.0461** (0.021)	−0.4808** (0.198)
High - Total Effect <i>P</i> -value	−0.0067 0.847	−0.0068 0.814	−0.0156 0.478	0.064 0.634
Observations	8703	8703	8703	8703
<i>R</i> -squared	0.027	0.024	0.018	0.023
Firms	2311	2311	2311	2311
Sample Avg	0.107	0.069	0.0992	0.267

in the past contribute to the market entry response. In Table 6 we further focus the analysis on a subsample of firms, for which we also observe product-level domestic sales data (all manufacturing firms with at least 10 employees), and analyze the role of new products in serving these new markets.

Column 1 of Table 5 demonstrates the most natural form of market entry: trade redirection, i.e., a situation

in which the export level of a given product to Muslim countries declines, while at the same time the export level of that same product to non-Muslim countries increases. The share of firms redirecting trade towards non-Muslim countries increases substantially and the magnitude corresponds to more than half of the sample average in earlier years. Interestingly, the trade redirection response is significant for both low- and high-leverage firms. At the

same time, high-leverage firms' response is smaller, consistent with other results and their financial constraints. Trade redirection is presumably the easiest and cheapest form of boosting revenues after the boycott, likely explaining why we observe high-leverage firms responding along that margin. A similar picture emerges from columns 2 and 3, which analyze the market entry response for product categories that have high and low share of new entrants among all sellers. Low-leverage firms enter both types of markets, although the coefficient is slightly higher for those with high share of new entrants (column 2). High-leverage firms, on the other hand, only show a positive response in product categories with above-average share of new entrants. More entry (and exit) in these markets suggests lower entry costs compared to other markets with a high share of stable exporters, which may in turn make entry easier for constrained firms after the boycott.

Columns 4 to 7 of Table 5 analyze how the product dimension of the market entry response relates to shape and evolution of a firm's past product portfolio. Specifically, we split previously exported HS-2 product categories into above/ below average level or growth of a firm's past exports, and report new market entry in each of these subsets of the past product portfolio. Responses for all product subgroups are significant for low-leverage firms, but are essentially zero for high-leverage firms. In terms of magnitudes, we find no difference by the size of HS-2 category in firm's past product portfolio: the market entry effect is similar for both above and below average HS-2 categories. In contrast, columns 6 and 7 show that the market entry response is two times as big for product categories in which a firm's past export growth was below average. This result is consistent with patterns documented earlier in Fig. 6 and suggests that market entry is most successful for markets with untapped demand for a firm's products.

Earlier results in columns 6 and 7 of Table 4 demonstrate that low-leverage firms increase the number of products that they export. That analysis, however, cannot distinguish between products that were sold domestically before but were not exported and products that were not produced at all. To do so, we now focus on a subset of manufacturing firms with at least 10 employees, for which we observe total (domestic and foreign) product-level sales with quarterly frequency. While the results of the analysis are based only on a subsample of all exporters, they can shed some light on the changes in firms' product portfolio overall, as opposed to their export portfolio only.

Consistent with the market entry response being partially driven by product expansion, Table 6 shows that low-leverage firms exposed to the boycott introduce differentially more new products in 2006, compared to other firms and to their past behavior. New product introduction is significantly less pronounced among exposed firms with high leverage and the total effect for this group is close to zero (columns 1 and 2). Contrasting the number of newly introduced products (column 4) with the number of newly introduced and newly exported products (column 5) reveals that 80 percent of newly introduced product are exported right away, and thus the modification of firms' product portfolios is primarily meant to enable the market entry response. The magnitude of changes for low-leverage firms

is large compared to new product introductions in a typical year, which is plausible given that the boycott requires major and urgent restructuring of the product portfolio. Yet, despite this significant adjustment, new product introduction accounts for a small share of the overall market entry response compared to the geographic expansion of existing products. Our calculations indicate that in 2006 15.3% of new export markets (defined as product-country pairs) involved a newly introduced product, and these markets represented 7.2% of export value to all new markets, consisted with new products being sold in smaller amount.

What drives the product portfolio adjustment that we observe? Given that we analyze one year after the boycott, and that technological innovation typically requires a long R&D process, the observed increase in the number of new products may be surprising. Additional analysis in Appendix Figure A.3 shows that these adjustments are significantly stronger for fast-moving industries (panels C–D), in particular textiles manufacturing (panel B), for which the product cycle is shorter and which are better able to quickly adapt to new conditions. Panels E and F further suggest that part of the effect is driven by exposed firms accelerating the introduction of new products that they had been developing; the quick response is especially pronounced in the category of “rising” products, i.e., those for which the share of new entrants among all sellers had been recently growing at above-average rate. Further analysis of the export data shows that new products introduced after the boycott often entail small adjustments and are closely technologically related (such as a producer of men's underwear may start selling men's pajamas or a machinery manufacturer selling replacement parts), helping to explain the fast response time of exposed firms. We also find firms starting to export complementary products (e.g., a firm exporting meat products adding sauces or spices), perhaps making use of their idle sales force and trying to bundle products to make them more attractive for their existing customers. Overall, the effects observed in our setting seem to utilize existing product pipelines and entail small modifications and adjustments that make the product more attractive to new, non-Muslim markets. These changes, while costly, can often be introduced right away, without significant time spent on the R&D process.

5.4. Spending and debt response

Entry into new markets and the modification of firms' product portfolio presumably require additional spending, and we expect difficulties in financing this additional spending to be the reason for the muted market entry response by high-leverage firms. To support this interpretation, we directly analyze whether high-leverage firms have lower levels of expenditures that can be associated with sales expansion, and whether they acquire less new financing. Specifically, columns 1 to 4 of Table 7 analyze the impact of the boycott on selling, general and administrative expenses, capital expenditures, and debt levels. Column 1 shows that SG&A expenses increase by over 6% for low-leverage firms. While the differential impact on high-leverage firms is not statistically significant, the total effect for high-leverage firms is also not significant and

Table 7

Spending and debt response.

All regressions are based on Eq. (3) and include firm fixed effects, industry-year fixed effects, and binary variables for each year interacted with indicators for high leverage. The main independent variable is the triple interaction of exposure to the boycott (exposed), having high leverage, and post-boycott period (year 2006). Dependent variables are the log of SG&A expenditures, log of 1 + total investment, log-changes in total debt, a binary indicator for a net increase of total debt, log-change in total wages and total number of full-time equivalent workers, as well as a binary indicator for using any outsourcing. The bottom row presents mean of the levels of dependent variables in the pre-boycott period in (monetary values in 1,000s of Danish Kroner). "High - Total Effect" presents the sum of baseline coefficient for low-leverage firms and differential effect for high-leverage firms and the *p*-value row presents the corresponding *p*-value from the *F*-test for significance of high-leverage firms' total response. In all regressions, standard errors are clustered at the industry level, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1) ln(SG&A Expend)	(2) ln(1+Inv)	(3) Δ ln(Debt)	(4) Debt Increase	(5) Δ ln(Wages) (Total)	(6) Δ ln(Emp) (FTE)	(7) Any Outsourcing
Exposed	0.0642*** (0.019)	0.3394*** (0.051)	0.0362* (0.018)	0.0602*** (0.009)	-0.0133 (0.016)	-0.0036 (0.021)	-0.0134 (0.016)
Exp X High	-0.0506 (0.031)	-0.2739*** (0.061)	-0.0634** (0.028)	-0.0434* (0.023)	-0.0377** (0.016)	-0.0503** (0.019)	0.0415* (0.021)
High - Total Effect (<i>P</i> -val)	0.0136 0.566	0.0655 0.279	-0.0272 0.164	0.0167 0.431	-0.0510*** < 0.001	-0.0539* 0.0876	0.0281** 0.0338
Observations	58,924	60,133	60,111	60,138	58,240	60,138	60,138
<i>R</i> -squared	0.021	0.059	0.013	0.014	0.010	0.016	0.022
Firms	13,951	13,963	13,962	13,963	13,684	13,963	13,963
Sample Avg	8108	2846	30,868	0.525	0.0406	0.0239	0.759

close to zero, suggesting that they did not see a meaningful increase in SG&A. The analysis of investment in column 2 similarly shows that low-leverage firms significantly increased capital expenditures and that the effect for high-leverage firms was significantly lower and indistinguishable from zero.

Consistent with the hypothesis that high-leverage firms do not expand and increase expenditures because of their inability to obtain financing, columns 3 to 4 of Table 7 show an increase in borrowing by low-leverage firms, but not by high-leverage firms. We find a 4% increase in total debt for low-leverage firms, that is significant at 10% level, and a highly statistically significant increase in the likelihood of a net increase in debt levels (column 4). In Appendix Table A.7 we present additional details of the debt response, demonstrating that the change is driven both by short-term debt to suppliers and debt to banks.²⁰ The differential response of high-leverage firms is negative and significant for both measures, leading to an insignificant total effect.

Lack of financial flexibility among high leverage firms may also explain an additional factor contributing to the differential product market and sales response: changes in receivables (Appendix Table A.8). The level of receivables increases for low-leverage firms (even though total sales

remain unchanged, Table 3), suggesting that low-leverage firms may be improving financing conditions offered to their customers and thus encouraging an increase in sales. As documented earlier in column 5 of Table 3, neither low- nor high-leverage firms significantly adjust prices after the boycott, even though the point estimate for low-leverage firms suggests that they may be lowering prices to a limited extent. Similar patterns are also observed in the more detailed analysis of price evolution in Appendix Table A.3. Point estimates suggest that low-leverage firms decrease their prices, while high-leverage firms do not, yet these estimates are not statistically significant. Overall, the lack of a statistically significant response and differences between the two groups points to a limited importance of price adjustments in response to the boycott, consistent with the aggregate evidence in Table 2.

Next, we analyze labor costs. While many cost categories (such as materials) may automatically decline following a decline in sales, labor contracts are typically not contingent on sales level, and thus their changes represent firms' active decisions regarding their cost policy. Columns 5, 6, and 7 of Table 7 demonstrate that low-leverage firms do not significantly change labor inputs, measured either by total payroll or total number of full-time equivalent workers. High-leverage firms, on the other hand, significantly reduce their employment, with both measures decreasing by around 5%. A similar decline of payroll and headcount suggests that the employment decline is not accompanied by a decrease in average wages. This is indeed confirmed by an additional direct analysis of average wages (Appendix Table A.9) and is consistent with the well-documented stickiness of wages, although may be somewhat surprising in the face of a clear negative shock.

While the decline in employment is consistent with general downsizing of the firm, one may expect that employment does not decrease after a temporary reduc-

²⁰ Note that these changes in liabilities are relative to high-leverage and low-leverage firms in the control group, thereby accounting for mean reversion among high and low leverage firms during normal times. In addition, we conduct a placebo test by defining high leverage based on debt holdings at the end of 2002 and analyzing changes in debt in 2004. This exercise aims to capture any differential regression to the mean among firms in the exposure group during normal times. Panel B of Appendix Table A.7 shows that, if anything, exposed low leverage firms reduce their liabilities, while high leverage firms increase them compared to firms in the control group. These differences would lead us to understate the effects in Table 7 and Panel A of Appendix Table A.7.

tion in sales. This stickiness, known as labor hoarding (Okun, 1963), is especially likely in our setting as firms face significant uncertainty about the length of the boycott. Hoarding labor, however, is costly. It requires paying workers' salaries today, even though they are not productive, and the benefits can only be recouped in the future in the form of reduced hiring and training costs. Given these costs, high-leverage firms may decide to reduce employment, especially of high-skill workers that are most expensive to hoard (which we confirm in Appendix Table A.9).

Interestingly, the decline in employment for high-leverage firms is accompanied by an increase in the use of outsourcing. Hence, the observed employment drop only partially represents a decline in labor inputs and reflects also changes in the nature of labor contracts: high-leverage firms reduce their use of more rigid employment contracts and increase their flexibility by outsourcing, which as a business-to-business contract is generally more flexible and more easily scalable. Column 7 of Table 7 shows that high-leverage firms are 2.8 pp. more likely to report any outsourcing after the boycott (the total effect for high-leverage firms, significant at 5% level, while the differential response is only significant at 10% level). This is a large effect given that 75% of firms before the boycott already declared using some outsourcing. A shift towards more flexible types of labor contracts is consistent with the theoretical model of Caggese and Cuñat (2008) and their empirical evidence that financially constrained firms are more likely to rely on fixed-term workers. Using data from a firm survey on purchased intermediate goods and services collected by Statistics Denmark, Appendix Table A.9 shows that newly outsourced activities are more likely to correspond to tasks performed by high-skill workers, e.g., lawyers, accountants, or IT specialists, which is consistent with the decrease in high-skilled workers' share in total employment reported in column 2 of Appendix Table A.9. Outsourcing high-skill services is further consistent with substituting non-core activities at these manufacturing firms: the results do not show any significant response in engineering services, suggesting that these firms remain focused on their core competencies.

Overall, the labor inputs response demonstrates that high-leverage firms do actively adjust to the boycott, and hence earlier null results for this group do not indicate lack of any reaction. However, high-leverage firms adjust in a significantly different way than low-leverage firms, focusing on cutting costs as opposed to recouping sales. The shift towards outsourcing also suggests that in the face of a negative shock, firms with high level of financial leverage try to reduce their overall risk by reducing operating leverage, i.e., shifting the costs to more variable basis, consistent with the tradeoff hypothesis (Van Horne, 1977).

5.5. Robustness

In this section, we describe several robustness checks that we use to augment the main analysis. We analyze the importance of the effects depending on the size of the shock and use alternative definitions of our main variables.

Appendix Table A.10 provides additional insights on the main effect by analyzing various measures of exposure to

the shock. We find robust support for our main findings using continuous measures of exposure and focusing only on firms that sold to Arab countries which declared a formal boycott. We also analyze the results for firms with particularly strong preexisting ties to Muslim markets. Specifically, we provide a subsample analysis for highly exposed firms with at least 10% of exports sold to Muslim markets in both 2004 and 2005 and compare them to less exposed firms. Consistent with our expectations, we obtain much larger negative effects on sales, debt, and employment for high-leverage firms with high exposure. This is consistent with the financial constraints mechanism: with larger exposure, the response alleviating the effects of the shock is likely to require higher capital outlays, and hence high leverage puts firms at a bigger disadvantage. Nonetheless, even for firms with lower exposure we find significant differences between low- and high-leverage firms in the product market response, indicating that product introduction may be especially sensitive to firms' ability to access capital.

We also conduct robustness tests for the main results using alternative measures of leverage. Specifically, we use financial leverage only, i.e., excluding debt to suppliers, we rank firms in high and low leverage within industries, and we conduct the analysis using continuous measures of book leverage or financial leverage. Appendix Table A.11 reports the results for our main outcomes of interest. In all specifications we obtain results similar to those from the main analysis. The results for the binary indicator of high financial leverage are less precisely estimated, but remain similar to all other specifications that confirm significant differences between low- and high-leverage firms. In Appendix Table A.12 we also study the role of short and long-term debt separately. The results show that along most outcomes both short- and long-term debt play a significant role. For some outcomes, such as entry into new non-Muslim markets and total exports to non-Muslim markets, the coefficient on the short-term debt is larger, which would be consistent with more urgent nature of constraints created by short-term liabilities. However, the difference between short- and long-term debt coefficients is never significant.

Finally, Appendix Table A.5 presents the results for the matched-sample analysis. Specifically, in the first step we match exposed and control firms using caliper matching on size (employment) and debt level. The analysis in Panel A reveals similar average effects of the boycott as reported in Table 2. In Panel B, we augment the match between exposed and control firms with second-step caliper matching on size between high- and low-leverage firms within the exposed group to ensure that the sample is balanced not only across exposure, but also across leverage. We again find quantitatively very similar results to the main findings in Tables 3, 4, and 7. These results further alleviate concerns about differences between exposed and control firms, and high- and low-leverage firms.

6. Discussion of the impact of leverage

How should we interpret the differential responses of high-leverage firms that we document in the context of

corporate finance theory, and in light of various possible explanations of observed patterns? In this section, we provide both theoretical and empirical arguments to support our interpretation of the results. We first discuss the potential mechanisms through which debt can affect adaptation to shocks. We conclude that our results are consistent with a class of models predicting that a higher level of debt (lower level of entrepreneur's initial equity) may lead to underinvestment. In the next step, we provide additional empirical results that support the financial constraints interpretation of our results. First, we use the share of debt maturing soon as an alternative source of variation in financing needs, and find very similar results as in our main specification that relies on leverage. Second, we entertain several potential alternative explanations and directly control for them in our regressions, confirming that the main results are not explained by other factors. Finally, we also analyze the medium-term impacts of the boycott.

6.1. Debt and firms' adaptation to shocks

Why would debt affect firms' response to a shock? Lower ability of high-leverage firms to expand in response to the boycott may be driven by credit rationing in the presence of asymmetric information (Jaffee and Russell, 1976; Stiglitz and Weiss, 1981). For any level of interest rate, an entrepreneur for whom the expansion is more risky will have higher benefits from applying for credit. That creates an adverse selection problem for the bank, which in response may not offer credit at any rate acceptable to the firm. This friction will make it impossible for firms with high level of debt to obtain financing for a risky expansion.

Another mechanism leading to underinvestment is debt overhang (Myers, 1977). If a firm cannot obtain financing from the bank that made the initial loan, other creditors may be hesitant to lend the additional money if the market value of existing debt is lower than its face value.²¹ Both asymmetric information and credit rationing limit a firm's ability to obtain financing only if the firm is unable to honor its debt obligations in case the risky expansion fails. Hence, these frictions will not affect firms for which initial debt levels are low, because even if the expansion fails, the debt can be repaid if initial equity is high enough.

The link between debt and underinvestment can also be related to shareholder preferences: more indebted firms are more likely to go bankrupt in case a risky project fails, and hence their shareholders may prefer less risky projects. More generally, debt may influence the trade-off between short- and long-term benefits and costs (Maksimovic and Titman, 1991). In our context, high-leverage firms choose downsizing and short-term savings by reducing their wage bill, and do not invest in long-term personnel retention or product introduction. These longer-

term strategies may be more risky, even though they may also be more profitable in expectation.²²

The overall pattern of our results is consistent with all of these models, and it is known that their implications are difficult to distinguish empirically (Froot et al., 1993; Stein, 1998). Yet, selected results allow us to speculate about the relative importance of various channels in our context. Debt overhang is most important when market value of existing debt is below its nominal value (Stein, 2003), which should result in bankruptcy of some firms. Yet, we do not find a significant increase in the probability of bankruptcy after the boycott. The shareholder preferences argument implies that high-leverage firms may find it optimal not to expand. Yet, an increase in trade redirection (Table 5) and the expansion in selected markets (Fig. 6) suggest that high-leverage firms try to expand to some degree, even though their response is muted relative to low-leverage firms.

We note that our results are inconsistent with other theories that generate the opposite relationship between debt and likelihood of expanding. For example, in the spirit of Jensen (1986), more indebted firms may be more willing to expand because the pressure of upcoming debt payments provides additional incentive to exert effort and try to recoup lost demand. Because debt payments have predetermined size, firms with large scheduled debt payments coming soon may prefer redirecting sales to cutting costs, since payments of a given size are easier to meet when a firm's scale is larger. Yet, our results clearly point in the opposite direction: high-leverage firms are less likely to invest in new market entry, and instead respond by reducing employment.

6.2. Differences in debt maturity

Our theoretical discussion above suggests that high leverage may impede firms' response to the boycott through the financial constraints channel. A potential concern with interpreting our empirical results through these lenses is that high leverage may also correlate with several omitted variables and thus its effect may capture something else than the effect of financial constraints. To alleviate this concern and support our interpretation, in this section we turn to the analysis of an alternative source of variation in financial constraints: timing of loans maturity around the boycott.

Our approach is similar to that taken by Almeida et al. (2011). Specifically, we use a yearly snapshot of all non-mortgage loans made to Danish firms by Danish banks to determine the share of long-term debt that matures soon after the boycott for each firm. The data has some limitations: mortgages and loans made by foreign banks are not included, the timing of the snapshot does not necessarily correspond to firms' fiscal year end, and original maturity of the loan is not available. Nonetheless, because we are able to observe the same loan over time,

²¹ In addition, in a richer model in which the risky expansion requires not just financial investment but also some additional input (e.g. effort of the entrepreneur), the debt overhang channel can also decrease investment of high-leverage firms via reduced incentives to provide this additional input, e.g., exert effort.

²² Several empirical papers also demonstrate that higher debt leads to lower focus on long-term outcomes, see for example Chevalier (1995); Matsa (2011); Phillips and Sertsios (2013); Cohn and Wardlaw (2016), and Kini et al. (2017).

Table 8

Debt maturity approach.

All regressions are based on Eq. (3) in which the measure of leverage is substituted with an indicator for high share of long-term debt maturing within 12 months after the boycott. All regressions include firm fixed effects, industry-year fixed effects, and binary variables for each year interacted with indicators for high leverage. The main independent variable is triple interaction of being exposed to the boycott (exposed), having high share of long-term debt maturing soon after the boycott, and post-boycott period (year 2006). Dependent variables are log-changes of the exports to Muslim countries, exports to non-Muslim countries (columns 1 and 2); 1 + number of new non-Muslim product country pairs (column 4), number of export destinations and export products (columns 5, 6), as well as total wages (column 7). In column 3, the dependent variable is a binary indicator for a net increase in debt compared to the previous year. The bottom row presents mean first differences (columns 1 and 2) and levels (columns 3–7) of dependent variables in the pre-boycott period. “High - Total Effect” presents the sum of baseline coefficient for unconstrained firms and differential effect for firms with the high share of debt maturing soon, and the *p*-value row presents the corresponding *p*-value from the *F*-test for significance of high-leverage firms’ total response. In all regressions, standard errors are clustered at the industry level, *** *p* < 0.01, ** *p* < 0.05, * *p* < 0.1.

	(1) Exposed firms	(2) Exposed firms	(3)	(4)	(5) All firms	(6)	(7)
	Δln(Export) (Muslim)	Δln(Export) (Non-Muslim)	Net Debt Increase	Δln(1+New N- Musl Prd-Ctr)	Δln(Export Countries)	Δln(Export Products)	Δln(Wages) (Total)
Exp X 2006	−0.5210*** (0.088)	0.1838*** (0.053)	0.0387*** (0.011)	0.1194*** (0.019)	−0.1442*** (0.018)	0.0685*** (0.014)	−0.0322** (0.014)
Exp X 2006 X Maturing Soon	0.0815 (0.133)	−0.2901*** (0.078)	−0.0238 (0.035)	−0.1356** (0.064)	−0.0819*** (0.030)	−0.1082*** (0.033)	−0.0700** (0.029)
High - Tot Effect (<i>P</i> -val)	−0.439*** 0.010	−0.106 0.164	0.0149 0.675	−0.0162 0.818	−0.226*** < 0.001	−0.0397 0.314	−0.102*** < 0.001
Observations	4790	6951	43,701	43,701	43,701	43,701	43,701
R-squared	0.026	0.009	0.009	0.035	0.032	0.036	0.010
Firms	1423	1626	12,526	12,526	12,526	12,526	12,526
Sample Avg	0.158	0.0396	0.517	10.90	7.832	11.38	0.0342

we are able to identify long-term loans, i.e., those present in the data for more than one year, and construct a measure of the share of long-term debt that will mature soon after the boycott. We define the independent variable, “Maturing Soon”, as an indicator for firms being above the 90th percentile in the distribution of share of long-term debt maturing within 12 months following the boycott. We regress key outcomes of our analysis on the interactions of Muslim exposure, post-boycott period, and the “Maturing Soon” variable. The specification is analogous to our main specification in Equation (3), except the measure of leverage is substituted with the measure of the share of debt maturing soon.

The results of this analysis, presented in Table 8, confirm the results from the main analysis presented throughout the paper. Column 1 shows that firms with debt maturing soon are hit by the same shock as other firms and lose the same share of exports to Muslim countries. Their ability to recoup lost sales by increasing exports elsewhere, however, again appears limited – while unconstrained firms increase their exports to non-Muslim markets by 20%, exports of constrained firms do not significantly change (column 2). Consistent with the financial constraints interpretation of these differences, column 3 indicates that unconstrained firms are significantly more likely to see a net increase in debt, but there is no significant change for constrained firms. Column 4 analyzes the main outcome of our analysis – market entry. Unconstrained firms enter significantly more new non-Muslim markets following the boycott, but the same effect is insignificant and close to zero for constrained firms. Decomposing the market entry responses by countries and products, we find a significantly larger decline in destination

countries and no increase in the number of exported product categories for more constrained firms, see columns 5 and 6, respectively. Finally, consistent with an active, yet differently focused response of constrained firms, column 7 demonstrates that the total payroll declines substantially more for firms with high share of debt maturing soon.

Overall, the results presented in Table 8 confirm the findings from our main analysis. Firms that happen to have a large share of debt maturing soon after the boycott, and are thus likely to be more financially constrained because they need to refinance their debt on top of gathering financing for potential expansion, behave very similarly to firms with high leverage, consistent with the financial constraints interpretation of our results. Appendix Table A.14 presents estimates from specifications that include controls for the level of leverage. The results remain very similar, consistent with a high share of debt that happens to mature right after the boycott being largely orthogonal to capital structure overall. We also explored an alternative definition of the threshold for the maturity variable, defining a broader constrained group as firms above the median share of debt maturing in the year after the boycott. This approach produces estimates that are always of the same signs as in Table 8, but the magnitude and precision of the estimates generally decrease, which is to be expected when including less constrained firms in the debt-maturing-soon group.

6.3. Alternative explanations

While exploiting differences in debt maturity helps to alleviate various concerns about endogeneity of capital structure, we aim to further strengthen our results by pre-

Table 9

Alternative explanations for differential adjustment by leverage.

Each panel in the table reports results for a different specification. All regressions are analogous to the main specification (3) but enriched with an interaction of exposure indicator, year fixed effects, and the additional regressor of interest. In panel A, the additional regressor is the number of products exported by the firm. In panel B we include managerial quality based on managers' average pay. In panel C, we control for pre-boycott volatility of sales. In panel D, the additional regressor is the log change in the number of export product-country pairs over 3 years before the boycott. The main independent variable is the triple interaction of exposure to the boycott (exposed), having high leverage, and post-boycott period (year 2006). Dependent variables are log-changes in exports to Muslim markets, total sales, an indicator for a net increase in total debt, as well as log changes in 1 + the number of new non-Muslim product-country pairs, number of export destination countries, number of exported products, and total wagebill. Notice that coefficients for Exposed X 2006 are not reported, because after additional regressors are added, there is no single coefficient that identifies the effect for low-leverage firms. In all regressions, standard errors are clustered at the industry level, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Model extension	(1) $\Delta \ln(\text{Export})$ (Muslim)	(2) $\Delta \ln(\text{Export})$ (Non-Muslim)	(3) Net Debt Increase	(4) $\Delta \ln(1+\text{New N-}$ $\text{Musl Prd-Ctr})$	(5) $\Delta \ln(\text{Export}$ $\text{Countries})$	(6) $\Delta \ln(\text{Export}$ $\text{Products})$	(7) $\Delta \ln(\text{Wages})$ (Total)
Panel A: Number of Export Products X Year FE X Exposed							
Exp X 2006	0.1073	-0.2755***	-0.0406*	-0.2203***	-0.0692***	-0.1154***	-0.0393**
X High Lev	(0.093)	(0.052)	(0.022)	(0.044)	(0.019)	(0.038)	(0.015)
Panel B: Manager Pay X Year FE X Exposed							
Exp X 2006	-0.0302	-0.2997***	0.0030	-0.2358***	-0.0523	-0.1497**	-0.0254*
X High Lev	(0.114)	(0.077)	(0.037)	(0.068)	(0.033)	(0.054)	(0.015)
Panel C: Sales Volatility X Year FE X Exposed							
Exp X 2006	0.1147	-0.2498***	-0.0419*	-0.2030***	-0.0625***	-0.1051***	-0.0353**
X High Lev	(0.095)	(0.048)	(0.023)	(0.046)	(0.017)	(0.038)	(0.015)
Panel D: Pre-Boycott Growth X Year FE X Exposed							
Exp X 2006	0.1144	-0.2365***	-0.0484*	-0.2060***	-0.0626**	-0.1067***	-0.0309*
X High Lev	(0.102)	(0.072)	(0.026)	(0.048)	(0.024)	(0.038)	(0.018)

senting additional evidence that addresses potential confounding factors directly. To alleviate the concern that the documented differences between low- and high-leverage firms represent some omitted factor correlated with leverage, we consider several alternative explanations.

One explanation for observed differences involves pre-boycott product portfolio: perhaps low-leverage firms have a richer product portfolio and hence it is easier for them to respond via expansion. Yet, panel A of Table 9 finds no evidence that the size of product portfolio can explain the effects of leverage.²³ It is also possible that low-leverage firms are more likely to expand because they are better managed. Excluding this concern may be difficult because characteristics of management are often unobserved. Thanks to our granular personnel data, however, we are able to identify managers' pay and education, and use it to proxy for quality. Panel B of Table 9 confirms that the effect of leverage remains very similar even if we control for Management Pay X Year FE X Boycott Exposure, while panel D in Appendix Table A.13 performs similar analysis controlling for manager education, alleviating concerns that manager pay proxies for entrenchment, rather than quality. The sample size of this analysis is smaller, because we can identify managers only for a subset of firms in our data. While this may explain lower significance of some estimates, the magnitudes of the coefficients change very

little. Furthermore, panel C shows that the results are unlikely to be explained by the fact that low-leverage firms have more volatile sales in general, and hence may be better prepared for demand shocks and better suited to redirect their sales. Related to that, panel I of Appendix Table A.13 shows that higher level of input adjustments before the boycott, as proxied by labor turnover, does not explain the fact that high-leverage firms reduce employment, rather than expand sales, after the boycott. Finally, panel D controls for the pre-boycott growth in the number of export markets. It may be the case that high-leverage firms expanded more before the boycott, and hence are less likely to expand after the shock. Yet, this possibility finds little support in the data, and pre-boycott differential growth does not seem to explain the effect of leverage.

Naturally, other alternative explanations are possible, and we further investigate some of them in Appendix Table A.13. We find no evidence that the effect of leverage may capture differences in production technology (as proxied by labor share, skill intensity, and asset tangibility), firm size, industry, firm age, or profitability. Overall, we find no evidence in favor of any of the alternative explanations, suggesting that our results are best explained by the direct impact of leverage and financial constraints.

6.4. Medium-term results

In our main results, we focus on the first year after the boycott, 2006. This choice corresponds to the duration of the boycott and avoids confounding effects of the onset of the financial crisis. In addition, it allows us to avoid

²³ Notice that the table presents only the triple-interaction coefficients. This is because after including interactions of exposure, year, and various firm characteristics, the effect for low-leverage firms is not identified by a single coefficient.

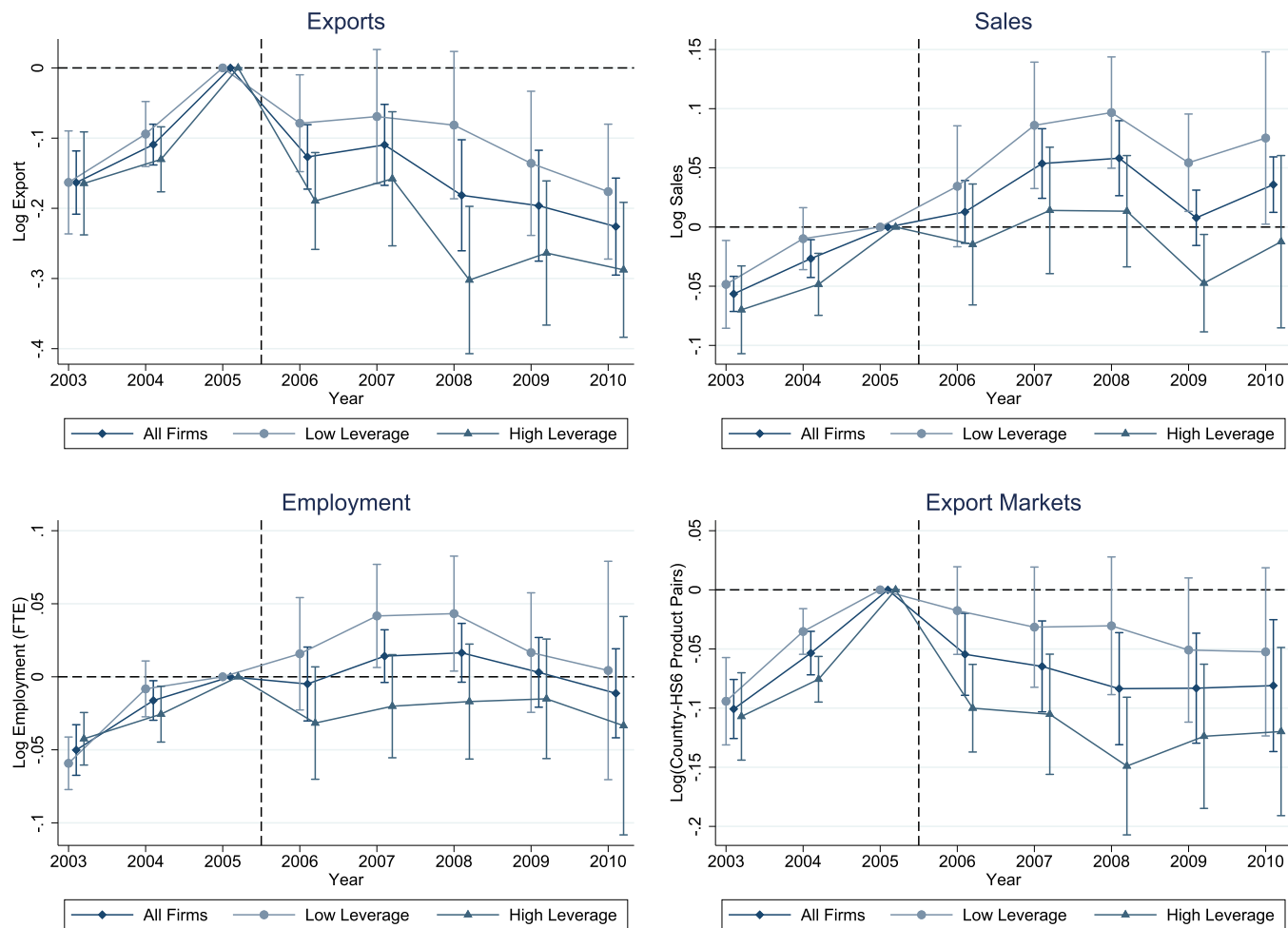


Fig. 7. Evolution of selected outcomes for low- and high-leverage firms until 2010. The markers in the figures show coefficients from event specification (2), where the sample was extended until 2010. Coefficients for low-leverage firms are for firms exposed to the boycott with leverage below median. Coefficients for high- versus low-leverage are differential effects for high-leverage firms within exposed firms. The brackets denote 90% confidence intervals.

a change in product classification that occurred in 2007 and may make estimates of the product-market response less reliable. Yet, it also means that we cannot draw definite conclusions about the optimality of different response strategies, because short-term profit indicators cannot provide reliable measures for the optimality of adjustment. In this section, we complement the results with a longer-run perspective, subject to the above-mentioned caveats of extending the analyzed period. Figure 7 presents the differential evolution of several key outcomes for firms exposed to the boycott up to 2010.

First, we find that the decline in exports appears persistent, although that is likely also related to a massive decline in international trade during the financial crisis. The initial difference between low- and high-leverage firms persists until 2010, although it is not typically significant. Sales for low-leverage firms remain higher over the entire period. Taken together, these results suggest that suffering a higher decline in sales during the boycott did not make high-leverage firms more resilient to the 2008 crisis. The impact on employment also appears to be rather persistent, at least before the crisis. The differences in export markets – which are defined as the number of product-country pairs to which a firm exports – persist as well, and actually widen in magnitude in later years. These results, however, should be interpreted with additional caution because of the change in product classification that occurred in 2007.

The long-term analysis is necessarily more speculative and less precise due to several considerations, with the onset of financial crisis being the most important one. Yet, what seems clear is that there is no evidence for a reversal of the initial patterns we document, and that at least some of the effects we document are longer-lasting. These findings support our interpretation of differences in adaptation being costly for high-leverage firms: while ex-ante it was possible that initial downsizing and lack of market entry expansion enable high-leverage firms to better weather the Great Recession, there is no evidence that supports this hypothesis.

7. Conclusion

This paper analyzes how firms adjust to an unexpected demand shock, and highlights the heterogeneous response by firms' capital structure. Our results underscore the importance of financial flexibility and, as a consequence, the importance of flexibility considerations in firms' capital structure decisions. Responding to the boycott, low-leverage firms are able to enter new markets and recoup lost sales to counteract the negative demand shock. For high-leverage firms, the expansion response is muted and total sales and employment decline.

We conclude with some limitations and comments on our study: First, we analyze a shock that significantly reduces revenues, but is too small to push firms to bankruptcy or drastically change the scale of their operations. An interesting complement of our analysis would be a study of a larger shock where firm survival and sizable adjustments are more relevant margins. On a similar note, our analysis studies a negative demand shock that affects

only a subset of markets in which the firm operates. A natural extension of our analysis would be a study of positive or broader demand shocks, as well as different types of environment changes, e.g., shocks to input availability or prices. While we expect that the reaction to different types of shocks would share many similarities with the effects we document, it may also involve some interesting differences. In particular, our results are likely to generalize to contexts in which heterogeneous changes in demand across market segments make entering new markets and modifying the product portfolio a viable response. For example, during the COVID-19 pandemic the optimal response for many firms might have involved shifting to remote delivery of services that were previously offered in person. Yet, it is clear that for some other type of shocks, e.g., overall decline in the availability of labor, the optimal response may be very different.

Second, an important missing element that determines the relationship between capital structure and flexibility is the ability for debt renegotiation. When a firm is hit by a negative shock and needs to flexibly adjust, are creditors willing to renegotiate their debt contracts? Data limitations prevent us from analyzing this question in this study, but more work is needed to extend existing evidence (Gilson et al., 1990).

Third, we note that our main results about firms' adaptation to this shock capture short-term responses within one year after the boycott started. While we do provide additional evidence about the impact of the boycott in later years in Section 6.4, the onset of the financial crisis and the great trade collapse make drawing definitive conclusions difficult. While we do not find evidence that high-leverage firms might have benefited from their post-boycott downsizing, we can not take a strong stand on the long-run costs or benefits of these differential adjustments. It is possible that financial distress forces firms to engage in necessary change (Jensen, 1986), which can benefit them in the long run. Boycott-driven structural change is confirmed by Friedrich (2022), who shows that a small share of exposed firms decide to de-layer and systematically change their occupational hierarchy, and internal wage structure.

Data availability

The code for this article can be found at <https://doi.org/10.17632/w9mg2vmz3z.1>.

Acknowledgments

We thank John Bai, Efraim Benmelech, Shane Corwin, Zhi Da, Mary Doheny, Sasha Indarte, Xavier Giroud, Vidhan Goyal, Timothy Loughran, David Matsa, Filippo Mezzanotti, John Shim, Yongxiang Wang, Jun Yang, Rafael Zambrana, and seminar and conference participants at Aarhus University, Northwestern University, University of Cologne, University of Notre Dame, University of Tuebingen, DIEW 2018, GEA 2019, MFA 2020, NFA 2020, and SED 2021 for their comments and suggestions. We thank the Labor Market Dynamics Group (LMDG), Department of Economics and Business, Aarhus University, and, in particular, Hen-

ning Bunzel for invaluable support and making the data available. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jfineco.2023.03.001.

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